



# Prospective Mitigation Interventions of Urban Heat Island in a Megacity: A Machine Learning Approach To Inform Climate Resilience in Dhaka, Bangladesh

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Received: 30 April 2025 / Revised: 4 September 2025 / Accepted: 5 September 2025  
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## Abstract

The Urban Heat Island (UHI) effect, characterized by higher temperatures in cities than rural areas, impacts the environment and the quality of life of city inhabitants. This study examines the spatiotemporal dynamics of the UHI effects in the megacity of Dhaka using Google Earth Engine remote sensing data and climate records. High-resolution Land Surface Temperature (LST) derived from Landsat-4/5 (TM), 7 (ETM+), and 8 (OLI\_TIRS) images spanning from 1990 to 2023 were analyzed to determine thermal trends across the Dhaka Metropolitan area. To evaluate the effects and effectiveness of UHI, various mitigation strategies, such as green roofs, urban forests, water conservation, high-albedo building materials, proper land-use planning, and population management, were considered. Normalized Difference Vegetation Index (NDVI), Normalized Difference Built-up Index (NDBI), Urban Index (UI), Normalized Difference Water Index (NDWI), and Albedo were estimated for different years, which were used for developing machine learning (ML) models to estimate LST. The relative performance of different ML models was examined to select the effective one for quantifying the impacts of mitigation strategies on reducing LST. The results showed a significant inverse relationship between LST and albedo, NDVI, and NDWI, and a direct relationship with UI and NDBI. The Support Vector Machine (SVM) model was the most effective in simulating LST compared to other ML models. Application of the SVM model revealed that a 10% reduction in NDBI could decrease LST by 15%. Increasing NDWI and NDVI by 20% can lower LST by 7% and 12%, respectively. While challenging, a 25% increase in NDVI has the potential to reduce LST by 15%. Although data and model uncertainties exist, this study offers pragmatic interventions to the planners and policymakers to significantly reduce UHI effects, improve living conditions, and build resilience against potential climate impacts in the rapidly growing Dhaka.

## Graphical Abstract

This graphical abstract visually summarizes the methodology, major findings, and proposed mitigation strategies for controlling the Urban Heat Island (UHI) effect in Greater Dhaka City. The upper section highlights the study's use of multi-temporal remote sensing data from Landsat satellites (TM, ETM+, and OLI/TIRS) spanning the period from 1990 to 2023. Key environmental indices—NDVI, NDBI, NDWI, Urban Index, and Albedo—were extracted using Google Earth Engine to understand urbanization-driven changes in land surface temperature (LST). The middle section presents the machine learning modeling process, where the Support Vector Machine (SVM) was identified as the best-performing model to predict

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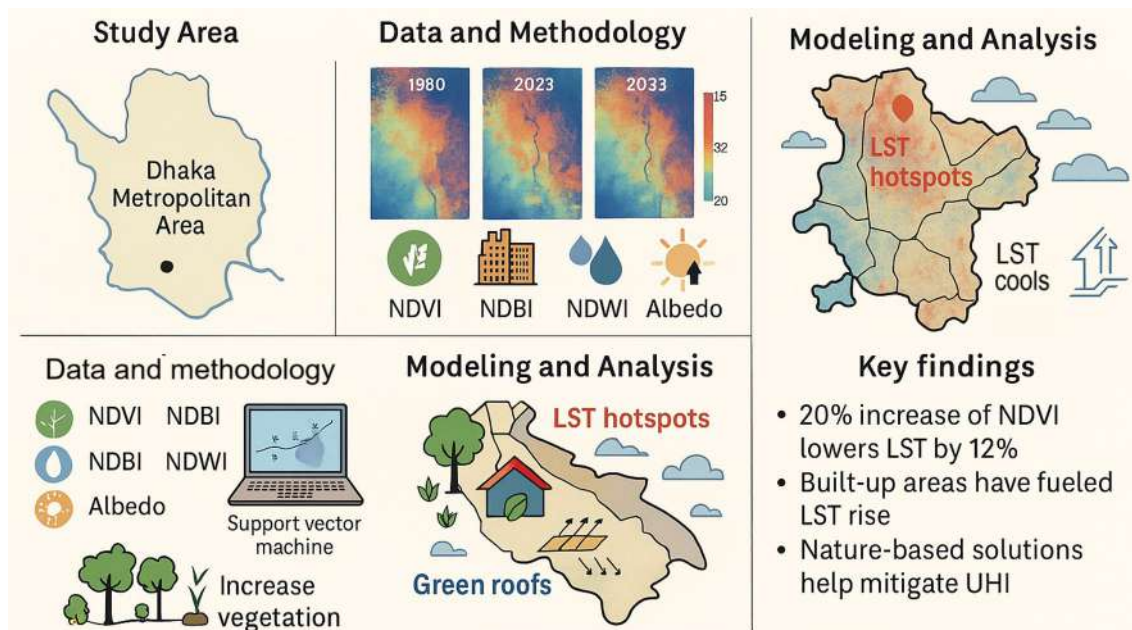
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LST patterns. Sensitivity analyses showed that built-up areas (high NDBI) are major contributors to elevating LST, while green spaces (high NDVI) and water bodies (high NDWI) are critical for cooling. Quantitative findings demonstrated that a 10% decrease in NDBI could lower LST by 15%, while a 20% increase in NDWI and NDVI could reduce LST by 7% and 12%, respectively. The lower section illustrates practical mitigation strategies recommended based on model results. These include enhancing green infrastructure (urban forestry, green roofs, vertical gardens), promoting water-sensitive urban design (conserving rivers, lakes, and wetlands), applying cool materials (high-albedo pavements and reflective roofs), and improving urban planning (integrated open spaces and climate-responsive land use management). Community engagement and sustainable transport initiatives are also emphasized as essential behavioral adaptations. Overall, the graphical abstract captures the comprehensive approach of the study, from historical LST trend analysis and machine learning-driven sensitivity evaluation to proposing localized, practical, and scalable solutions. It highlights how targeted environmental management can mitigate UHI intensity, enhance urban livability, and contribute to the climate resilience of one of the world's fastest-growing megacities.



## Highlights

- Analyzed UHI trends in Dhaka from 1990 to 2023 using Landsat data and environmental indices.
- Urban water bodies, vegetation cover, and high-reflecting building materials can reduce the UHI effects significantly.
- Increasing NDVI by 25% could reduce LST by up to 15% in Greater Dhaka.
- A 20% rise in NDWI can lower LST by approximately 7%, aiding urban cooling.
- New industrial zones contributed significantly to the formation of emerging LST hotspots.

**Keywords** Landsat images · Land surface temperature · Normalized predictors · Support vector machine · Sensitivity analysis

## 1 Introduction

Urbanization transforms regions into more developed, less green, and densely populated areas, leading to significant environmental changes. Such rapid urban expansion, characterized by the increase in urban residents, contributes to microclimate alterations, air pollution, and the loss of vegetation, often involving deforestation (Pour et al. 2020). One critical consequence of urbanization is the Urban Heat

Island (UHI) effect, where temperatures in urban cores are higher than in surrounding rural areas due to increased human activity and infrastructure.

The megacity Dhaka, one of the fastest-growing unplanned cities in the world, is experiencing UHI issues acutely. The elimination of green spaces and water bodies has intensified the UHI effect, leading to significant temperature increases and adversely affecting the city's microclimate, public health, and overall quality of life (Nuruzzaman

2015; Tashnim and Anwar 2016). The Greater Dhaka area, including key regions such as Dhaka North and South City Corporations, Gazipur, Tongi, Savar, Narayanganj, and Keraniganj, has recorded temperature rises of up to 6 °C compared to adjacent rural areas due to this phenomenon. Trotter et al. (2017) demonstrated that land surface temperature (LST) remains relatively stable over vegetated and water-covered surfaces but tends to rise significantly in areas with impervious surfaces, such as built-up urban regions, which cause losses of natural land covers, such as wetlands and agricultural lands, in Dhaka city. This urban growth exacerbates UHI and contributes to the formation of “hot islands,” where LST increased by as much as 7 °C over a decade (Ashikur et al. 2020). Koutroumanou et al. (2025) demonstrated that densely built zones in Nicosia, Cyprus, exhibit higher nighttime temperatures and lower humidity compared to cooler sparsely built zones, underscoring the importance of localized urban modeling for effective climate change mitigation.

In a densely populated city like Dhaka, several studies have assessed the impact of UHI. Uddin et al. (2022) demonstrated that rapid urbanization has significantly increased LST, closely linked to changes in land use and land cover (LULC). Tabassum et al. (2025) analyzed satellite data in Chittagong and found that urban sprawl has intensified the UHI effect. Studies showed that the expansion of built-up areas at the cost of green spaces and water bodies has exacerbated global warming. Kafy (2021) used Landsat imagery from 2000 to 2020 to show that urbanization in Dhaka increased LST by 7.24 °C, with projections indicating a 13% decline in green cover and a 21% rise in built-up areas by 2030. However, these studies primarily focus on UHI trends and urbanization, without examining the effectiveness of mitigation strategies in Dhaka.

Globally, UHI mitigation has been explored through the application of 3D models, building footprints, and satellite data (Rahman et al. 2022a, b), which highlight the value of high-albedo materials in reducing surface temperatures. Several approaches—such as green roofs, urban forestry, pervious pavements, and conservation of water bodies—have been identified, although each method comes with limitations (Nuruzzaman 2015). Integrated water and stormwater management are also central to sustainable urban design. For instance, Karamoutsou et al. (2024) emphasize sustainable urban stormwater strategies that reduce runoff and mitigate urban heat impacts, while Leivadiotis et al. (2024) demonstrate how remote sensing can be used to evaluate extreme events linking climatic extremes to urban resilience challenges.

Traditional knowledge also plays an important role. Fadhillah et al. (2023) highlight the effectiveness of community-based disaster management (CBDMD) and remote sensing

for early fire detection, while Vaulina et al. (2024) discuss sustainable land practices in coconut farming that balance ecology and economics. These examples show the benefits of combining local knowledge with modern tools. Modeling techniques such as multiple linear regression (MLR), artificial neural networks (ANN), and Feed-Forward Deep Neural Networks (FF-DNNs) have been used to predict LST and evaluate mitigation strategies, with studies showing that green space expansion can help lower urban air temperatures, even though the benefits may be constrained by continued urbanization (Maheng et al. 2019). Karamoutsou and Psilovikos (2021) also demonstrate the value of deep learning in water resource management through a case study of Kastoria Lake in Greece, highlighting its potential for informed environmental decision-making.

Environmental indices such as the Normalized Difference Vegetation Index (NDVI), Normalized Difference Built-up Index (NDBI), and Normalized Difference Water Index (NDWI) have further elucidated UHI dynamics. Studies have shown that LST is negatively correlated with NDVI and NDWI, and positively correlated with NDBI, highlighting the importance of vegetation and water bodies in mitigating UHI and enhancing climate resilience (Mirchooli et al. 2020; Karamoutsou et al. 2024). Furthermore, remote sensing, combined with nonlinear modeling, has proven effective for ecosystem and drought monitoring. Alibakhshi et al. (2024) investigated water and energy dynamics in the Kobi Wetland in Iran, providing a strong example of how advanced modeling techniques can offer actionable insights for ecosystem management and climate adaptation.

Despite extensive research documenting urbanization-driven increases in LST and UHI effect in Dhaka (Uddin et al. 2022; Kafy 2021; Trotter et al. 2017), significant gaps remain. Previous studies have primarily focused on detecting LST changes and associating them with LULC transitions, but have largely overlooked the evaluation of mitigation strategies to control UHI (Nuruzzaman 2015; Tashnim and Anwar 2016). Leivadiotis et al. (2024) evaluated the use of satellite data for hazard assessment, such as flood and UHI, as a key strength in their study, demonstrating how technology can aid in disaster risk management. Moreover, most research has relied on traditional statistical methods over relatively short time frames, without employing advanced predictive models such as machine learning or conducting long-term, high-resolution spatial analyses (Maheng et al. 2019; Pour et al. 2020). Furthermore, comprehensive analyses integrating multiple biophysical indices, such as NDVI, NDBI, NDWI, UI, and Albedo, with LST have been limited (Mirchooli et al. 2020). Despite the growing global emphasis on UHI mitigation strategies, few

studies have translated their findings into actionable recommendations for urban planners and policymakers (Rahman et al. 2022a, b).

Addressing these gaps, this study applies machine learning techniques to multi-decadal satellite data to assess spatial-temporal UHI patterns, explores the controlling influence of multiple environmental indices, and proposes targeted mitigation strategies to inform sustainable urban development in Dhaka.

This study aims to identify and evaluate effective mitigation measures for controlling UHI in Dhaka and inform urban planning and policy decisions to reduce the UHI effect and improve the urban climate. This study utilizes high-resolution satellite data from Landsat-4/5 (TM), Landsat 7 (ETM+), and Landsat 8 (OLI\_TIRS) from 1990 to 2023 to examine LST variations in Dhaka. By applying machine learning models and linear regression techniques, the study explores the relationships between LST and indices such as the NDVI, NDBI, UI, NDWI, and Albedo. This research offers valuable insights into the spatial distribution

of UHI in Dhaka, highlighting hot and cool zones, and proposes practical strategies for mitigation. The study's results are crucial for guiding future urban development in Dhaka, ensuring a sustainable balance between growth and environmental preservation.

## 2 Study Area

Dhaka, the capital of Bangladesh, with 8.9 million residents within its city boundaries and over 21 million people, ranks as the ninth-largest metropolitan area and the 6th most densely populated city globally. It is the largest city in Eastern South Asia and is the country's cultural, political, and economic hub. The Dhaka Metropolitan Area comprises several suburban regions, including the Dhaka South City Corporation, Dhaka North City Corporation, Savar Upazila, Keraniganj Upazila, Gazipur District, Narayanganj District, and Purbachal, as shown in Fig. 1.

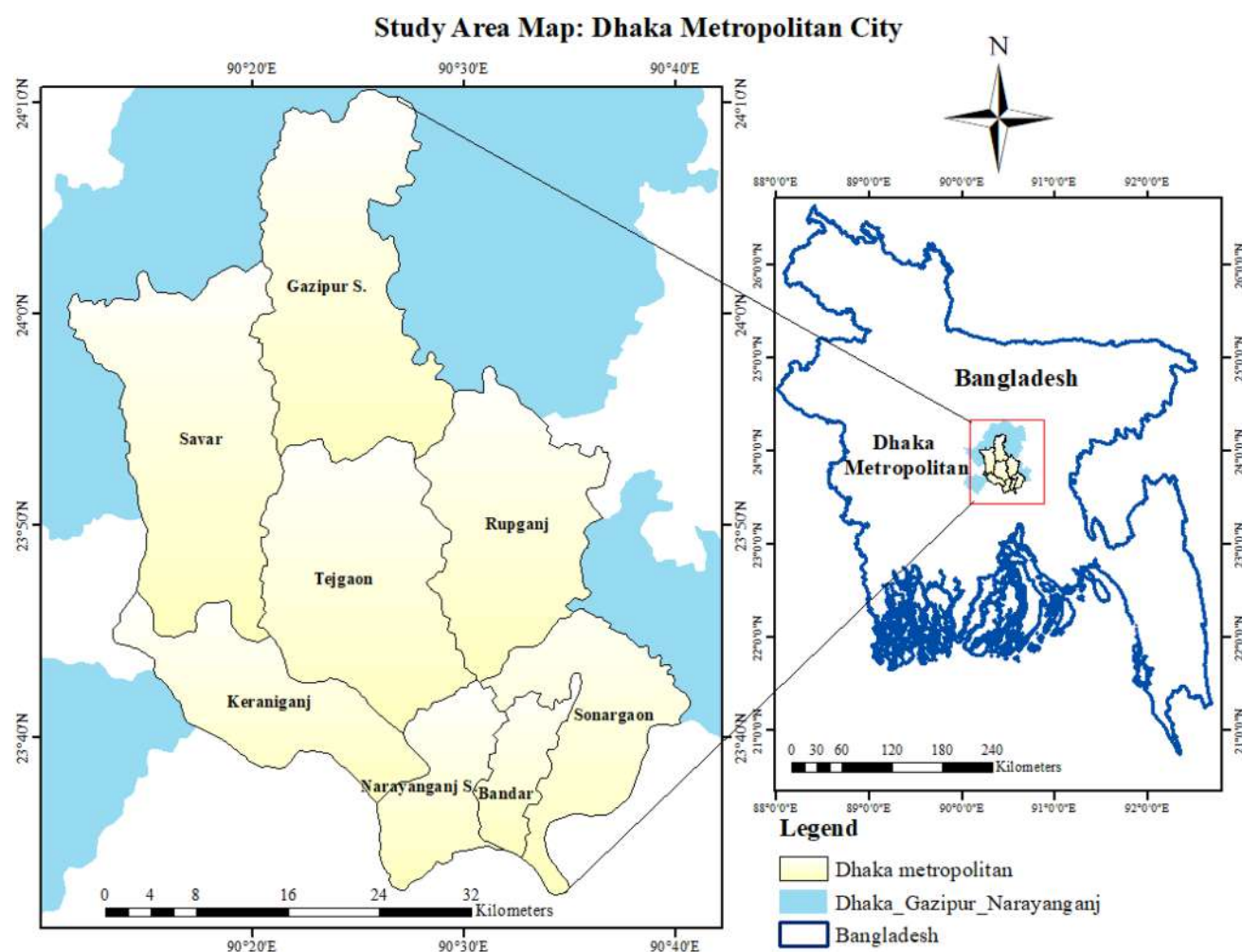


Fig. 1 Study area map of Dhaka Metropolitan area

Situated 32 m (104.96 feet) above sea level, the Dhaka City Corporation occupied 360 km<sup>2</sup> within the 1600 km<sup>2</sup> Dhaka Metropolitan Area (Choudhury 2007). The city's rapid urbanization has led to UHI effects due to increased density and the loss of green spaces (Begum et al. 2021). Given its economic significance and high population density, addressing UHI effects in Dhaka is crucial for the country's development. Modeling can estimate temperature changes based on LST indicators, underscoring the need for targeted UHI mitigation strategies in this rapidly growing area.

### 3 Data and Sources

Table 1 provides the data used in this study and their sources. The district shapefile is also available on the DIVA-GIS website. Other data can be downloaded using Google Earth Engine (GEE).

### 4 Methods

The procedures followed in the study to achieve its objectives are illustrated pictorially in Fig. 2. Details of the methods are given in the following sections. ArcGIS 10.4.2,

**Table 1** List of data used in this study and their sources

| Data           | Data Type                         | Time                               | Sources                                                                                                                                            |
|----------------|-----------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Shapefiles     | District, Upazila, Study Area     | Recent/Updated                     | Base map, DIVA-GIS<br>The Study Area is clipped in ArcGIS                                                                                          |
| Google Imagery | Google Earth Pro                  | Historical and Recent              | Google Earth Pro                                                                                                                                   |
| Landsat-5      | LST, NDVI, NDBI, NDWI, UI         | 1990, 1993, 1996, 1999             | Duration: March 1, 1984–June 5, 2013<br>It was equipped with the instruments Thematic Mapper (TM) and Multispectral Scanner (MSS).                 |
| Landsat-7      | LST, NDVI, NDBI, NDWI, UI         | 2002, 2005, 2008, 2011             | Duration: April 15, 1999–April 6, 2022<br>USGS continued scientific data collection by the Enhanced Thematic Mapper + (ETM+).                      |
| Landsat-8      | LST, NDVI, NDBI, NDWI, UI, Albedo | 2014, 2017, 2020, 2023 (2014–2023) | Landsat 8 has two sensors: the Operational Land Imager (OLI) and the Thermal InfraRed Sensor (TIRS).<br>Landsat 9 is a rebuild of its predecessor. |

SNAP, GEE, Google Earth Pro, and R scripts with relevant statistical packages were used to implement these methods. The study is conducted in two broad steps: (1) generating high-resolution UHI maps of Dhaka city, and (2) identifying measures for reducing the UHI of Dhaka city and conducting a modeling simulation. The equations for calculating NDVI, NDWI, NDBI, UI, and Albedo are well established in the literature and have been widely applied in previous studies (e.g., Xi et al. 2019; Kshetri 2018). In this study, these indices were derived using Google Earth Engine (GEE), where the respective formulas are implemented, and the processed outputs were subsequently downloaded for further analysis. Detail of the procedure is given in the following section.

#### 4.1 UHI Map Generation

The following steps are used to generate UHI maps from Landsat data using GEE:

- Calculate the NDVI using Landsat images of red and NIR bands.
- Produce Surface reflectance (SR) data from Landsat 4/5 and 8 using the Land Surface Reflectance Code (LaSRC) algorithm.
- To estimate cloud coverage and shadowing, use the USGS quality assessment band (BQA) through GEE.
- Estimate surface emissivity values using the LST retrieval technique from the ASTER GEDv3 dataset.
- Implement a vegetation correction based on mean ASTER GEDv3 NDVI and NDVI generated from Landsat to account for fractional vegetation cover (FVC) (Kshetri 2018):

$$FVC = \left( \frac{NDVI - NDVI_{bare}}{NDVI_{veg} - NDVI_{bare}} \right)^2 \quad (1)$$

Above are the NDVI values of fully vegetated pixels (NDVI<sub>veg</sub>) and completely bare pixels (NDVI<sub>bare</sub>).

- Calculate the emissivity values across vegetated areas at any given period using the vegetation-cover technique (Kshetri 2018).

$$\epsilon_b = FVC \epsilon_{b,veg} + (1 - FVC) \epsilon_{b,bare} \quad (2)$$

where bare represents the emissivity of vegetation and bare land, respectively, for a particular spectral band.

- Obtain LST data from Meteosat LST Climate Data (MFG) using linear regression using the statistical



**Fig. 2** Methodological framework showing the procedure in the study

mono window (SMW) algorithm. This model includes surface emissivity in a linearized radiative transmission equation (Ermida et al. 2020):

$$LST = A_i \frac{Tb}{\epsilon} + B_i \frac{1}{\epsilon} + C_i \quad (3)$$

where  $\epsilon$  is the channel's surface emissivity coefficient and  $Tb$  is the TIR channel's TOA brightness temperature. The algorithm's  $A_i$ ,  $B_i$ , and  $C_i$  coefficients are calculated using radiative transfer regression equations (Ermida et al. 2020).

viii. A graph is constructed from two cross-sectional profiles on LST and indicator maps to estimate UHI and the low

temperatures surrounding the city. Extracting profile CSV files helps examine trends.

ix. The LST, NDVI, NDBI, NDWI, UI, and Albedo maps have been created for several years to track UHI sites and the impact of indicators. Maps are made in ArcGIS 10.4.1.

The second step was to list the different mitigation measures adopted in cities. This material came from online journals and research manuals. The term “Mitigation of Urban Heat Island” was used to find articles on Google Scholar and Scopus. Only high-impact, high-quality journals were chosen. The selected journals' abstracts were carefully evaluated to ensure relevance to this research.

## 4.2 Critical Review of Literature for Mitigation Measures

A critical review of global UHI mitigation measures was conducted to extract and synthesize adaptation strategies across sectors, assessing their effectiveness in improving urban environments. For Dhaka, the most suitable measures were selected using four criteria: relevance (meaningful for Dhaka's context), comprehensibility (clear to all stakeholders), controllability (actionable through policy or practice), and sensitivity (responsive to environmental change).

## 4.3 Field Observation and Validation

Ground-based real-time LST monitoring stations are currently not available in Bangladesh, which limits the ability to conduct in-situ validation. Additionally, temperature measurements using thermal cameras, surface thermometers, or handheld thermal sensors were not utilized during fieldwork due to timing constraints. Since corresponding ground-based average temperature data for the same periods are unavailable, validation using field-based thermal measurements was not feasible. Consequently, this study relies solely on satellite-derived LST data. However, field observations were conducted to verify land cover classifications and confirm the spatial accuracy of identified hot and cool spots. Hotspot areas identified from satellite imagery generally correspond to urbanized zones, while cool spots align with non-urban or vegetated areas, validating the general spatial trends observed in the LST maps.

## 4.4 Model Generation

Randomly selecting 10,000 metropolitan map spots is used to simulate Dhaka's LST. The places were selected to cover the entire city. The extracted data is used to model LST using linear regression and ML model with NDVI, NDWI, NDBI, and UI as inputs.

### 4.4.1 Multiple Linear Regression Model (LM)

LM derives a relationship between an input variable and one or more outcome factors. According to Hastie et al. (2009) and Hastie and Tibshirani (2017), estimating coefficients that minimize the sum of squared errors (SSE) involves fitting a linear model to the data in linear regression. The equation for linear regression with many variables is:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + \varepsilon \quad (4)$$

The equation describes the relationship between the target variable  $y$  and the variables  $x_1, x_2, \dots, x_n$ . The variables

$x_n$  are the input variables,  $\beta_0, \beta_1, \dots, \beta_n$  are the coefficients that need to be estimated, and  $\varepsilon$  is the error term.

### 4.4.2 Machine Learning Methods

**4.4.2.1 Artificial Neural Network (ANN)** Artificial neural networks (ANNs) are computational models inspired by the brain's interconnected neurons, comprising multiple layers of linked nodes that learn complex, non-linear relationships. The ANN-based workflow in this study followed three phases: selecting an appropriate training procedure, identifying optimal neuron and layer configurations, and rigorously evaluating and validating model performance (Samuel et al. 2021); conceptually, information is processed through interconnected "neurons" analogous to biological networks (Srivastav and Bank 2021). In line with these principles, this study implemented a convolutional neural network (CNN). This specialized ANN architecture leverages local receptive fields and parameter sharing to capture spatial patterns and interactions in gridded urban and climatic predictors. This choice is well-suited to modeling intricate urban–climate relationships and to learning generalizable signals from temporal variability, including fluctuations in the urban heat island (Wilcke et al. 2013; Zhao et al. 2023).

**4.4.2.2 Random Forest (RF)** RF builds a model from bootstrap samples using a decision tree. Due to bootstrap sampling, RF is less likely to overfit than decision trees, even if they divide the training data (Wilcke et al. 2013). RF's ability to process high-dimensional data and manage interactions among numerous predictors makes it especially suitable for modeling the spatial complexity of urban heat patterns. A key advantage of RF is its ability to evaluate feature importance, which helps identify critical UHI drivers, such as vegetation cover, impervious surfaces, and proximity to water bodies. Its resilience in noisy, multi-variable environments makes it a strong candidate for UHI modeling in rapidly urbanizing regions, such as Dhaka (Rahman et al. 2021).

**4.4.2.3 Support Vector Machine (SVM)** SVM develops a regression model by matching training data and maximizing error tolerance (Vapnik and Cortes 1995). Significant margins separate SVM hyperplanes from data points. Support vectors, also known as indispensable hyperplane data points, alter the model's prediction function. By using kernel functions, SVM can map input data into higher-dimensional spaces, allowing it to capture complex, nonlinear relationships that are common in urban climate dynamics (Pour et al. 2018). This makes SVM particularly well-suited for

UHI modeling, where a combination of spatial, temporal, and environmental variables influences land surface temperature. Its strength lies in handling high-dimensional data from remote sensing and accurately modeling interactions across diverse predictors such as vegetation, impervious surfaces, elevation, and seasonal variation (Li et al. 2023).

The accuracy of the models was evaluated using standard performance metrics, including Root Mean Square Error (RMSE), Percent Bias (PBIAS), and the Kling-Gupta Efficiency (KGE). SVM, ANN, and RF can capture the key nonlinear relationships and spatial variability associated with urban thermal dynamics (Noi et al., 2017; Seto et al. 2021; Rahman et al., 2021; Li et al. 2023; Zhao et al. 2023).

**4.4.2.4 Model's Hyperparameter Tuning** Hyperparameters for the machine learning models were optimized using a grid search approach combined with 10-fold cross-validation to prevent overfitting and ensure generalization (Mirchooli et al. 2020). The optimal set of parameters was selected based on the lowest root mean square error (RMSE) and the highest  $R^2$  value on the validation data. The optimized hyperparameter values of the models are provided in Supplementary Table S1. For the CNN, three convolutional blocks with filters 32/64/128 and  $3 \times 3$  kernels were followed by a dense head with 128 units and dropout 0.30; L2 regularization was  $1 \times 10^{-4}$  on all trainable layers; optimization used Adam with an initial learning rate of  $1 \times 10^{-3}$ . For the RF, this study used 500 trees, maximum depth 20, minimum samples per split 4, and minimum samples per leaf 2. For the SVM (with RBF kernel), the optimal settings were  $C=10$ ,  $\gamma=0.01$ ,  $\epsilon=0.10$ , shrinking enabled, and tolerance  $1 \times 10^{-3}$ .

## 4.5 Sensitivity Analysis

This study estimated the sensitivity of five meteorological variables (NDVI, NDWI, NDBI, UI, and Albedo) on LST. The global sensitivity analysis method, developed by Sobol (1993, 2001), was used to achieve this goal. Sobol's method employs a variance decomposition approach to determine the sensitivity indexes of various variables. This study assessed climate variable sensitivity using a  $\pm 25\%$  test variance range. The method developed by Sobol and made available in the Sensitivity package for R (<https://cran.rproject.org/web/packages/sensitivity/index.html>) was utilized.

## 5 Results

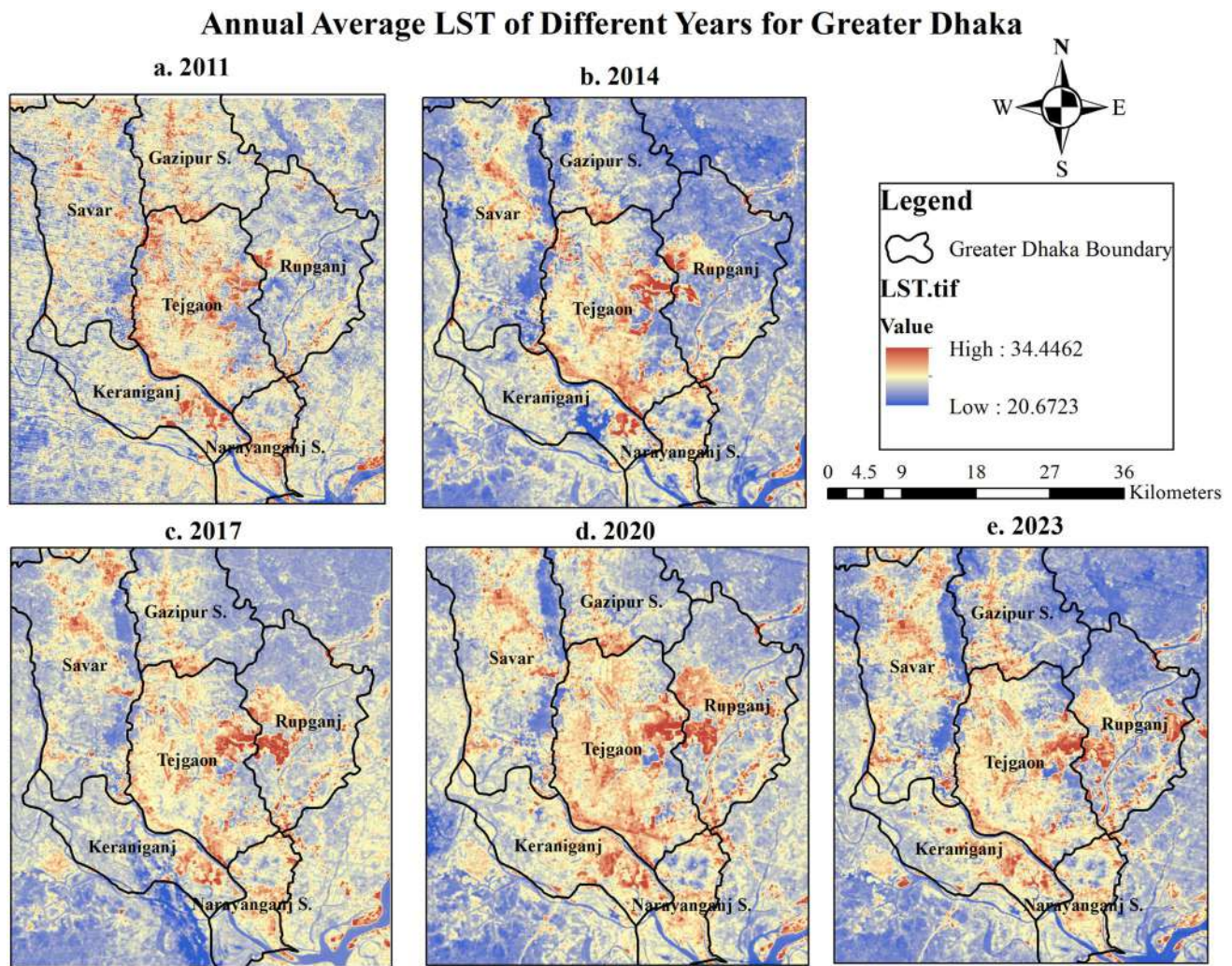
### 5.1 Spatial Distribution of Annual LST

Figure 3 illustrates the changes in LST of Dhaka city from 2011 to 2023, showing a progressive increase in temperature across the region. The analysis highlights an apparent rise in temperature within the urban core, indicating the intensification of the UHI effect as Dhaka's metropolitan area (DMA) expands. The imagery reveals a clear emergence and expansion of high-temperature zones over time, particularly in rapidly urbanizing peripheral areas such as Keraniganj, Purbachal New Model Town, and Narayanganj. Notably, several hotspots visible in 2023 were absent in 2011, indicating significant microclimatic changes linked to recent urban development. The temporal sequence of maps also shows a consistent rise in surface temperatures across earlier years, reinforcing the observed trend. This upward trajectory reflects not only the broader impacts of global climate change but also localized factors such as increased impervious surface coverage, vegetation loss, and intensified land use. Collectively, the results demonstrate a substantial growth in both the density and spatial extent of high-LST zones across the Dhaka region.

Supplementary Figures S-1 to S-3 illustrate seasonal LST variations in Dhaka city from 2013 to 2023, covering winter, pre-monsoon, and post-monsoon periods. The results show a consistent rise in LST across all seasons, indicating an intensifying UHI effect driven by urban expansion. The maps highlight the emergence of new urban heat zones, particularly in recently developed or rapidly urbanizing areas. Pre-monsoon LST increases are the most pronounced, likely due to extensive vegetation loss and shrinking water bodies, which significantly reduce the city's natural cooling capacity during this season. In contrast, winter maps reveal a more distinct UHI pattern, with urban cores exhibiting notably higher temperatures than their rural surroundings. Monsoon LST could not be assessed due to the lack of satellite data during that period. These seasonal patterns collectively underscore the strong influence of land cover change on Dhaka's microclimate and reinforce the need for targeted, season-specific mitigation strategies.

### 5.2 LST Profiles and UHI Mapping

UHI profiles for 2018, and 2023 are in Fig. 4. It shows high temperatures as red and low temperatures as blue. Profile-1 explores Dhaka's highest temperature zone, from southwest to northeast. Cities have the highest temperatures and fade to vegetated, peri-urban, and non-urban environments. Dhaka city is hotter than the surrounding area, as seen by Profile 2, which runs from southeast to northwest through

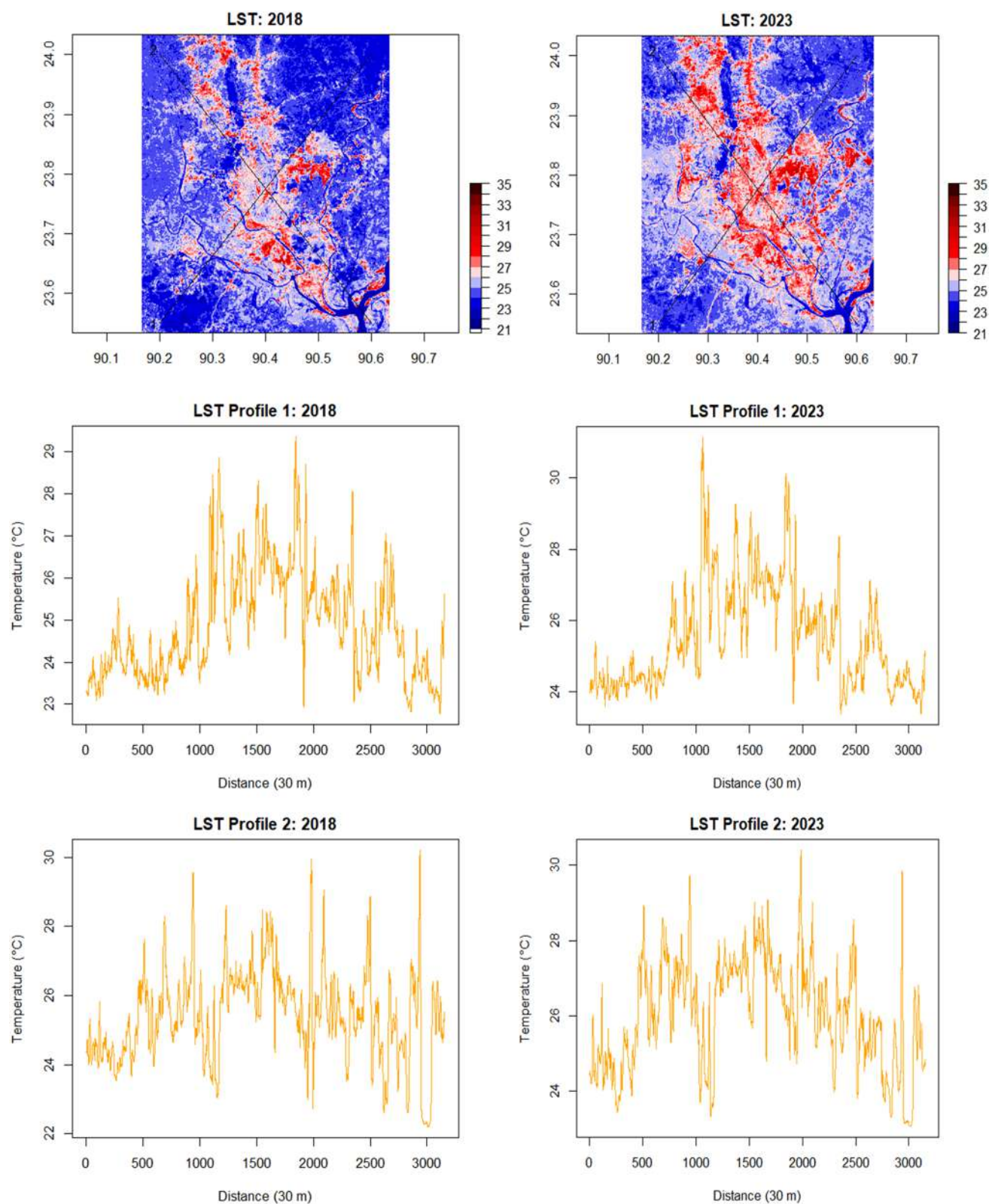


**Fig. 3** LST of Different years showing the high temperature in City centers (a) Annual average LST of 2011; (b) Annual average LST of 2014, (c) Annual average LST of 2017, (d) Annual average LST of 2020, and (e) Annual average LST of 2023

urban and vegetated areas. By 2023, some newly developed or peri-urban areas had high temperatures, and most Dhaka now had substantially higher temperatures. The warmest temperature was 30 °C in 2023, up from 29 °C in 2018. The red patch is larger than the blue one and signifies a higher temperature. This growth reflects the intensifying UHI effect driven by rapid urbanization across the Dhaka Metropolitan Area. The emergence of new hotspots in urban expansion zones further confirms the strong correlation between land development and increased surface temperatures. Seasonal UHI profiles, including the pre-monsoon curves shown in Supplementary Figure S-4, also reveal a significant rise in city temperatures, reinforcing the observed trend of UHI intensification throughout the year.

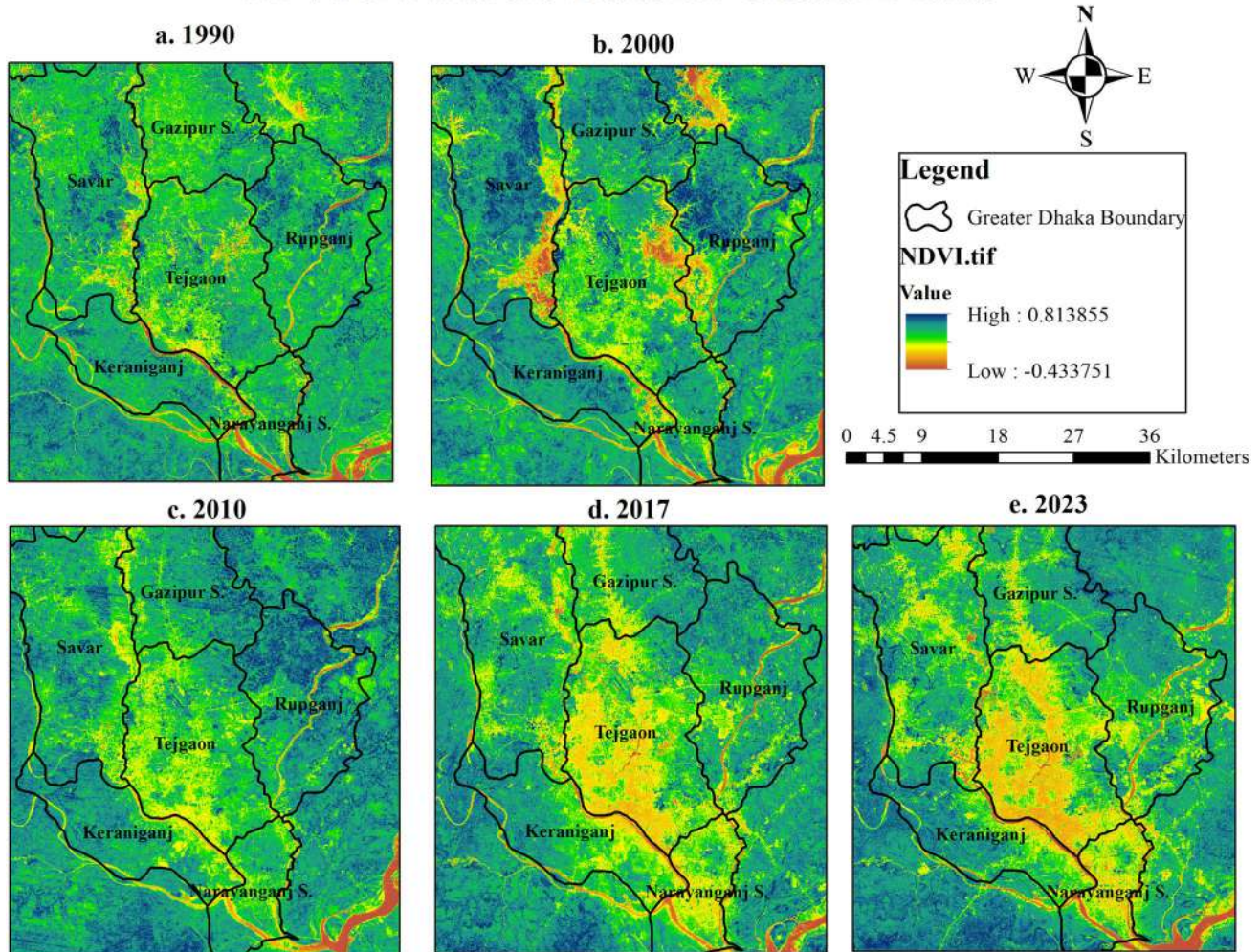
### 5.3 Indicators Mapping of Different Years

Figure 5 shows the spatial distribution of annual NDVI for 1990, 2000, 2010, 2017, and 2023. Here, yellow to red denotes high NDVI levels, whereas blue-green suggests high NDVI values. The photographs show a substantial loss of vegetation inside Dhaka's urban zone. The NDVI has decreased as the metropolitan region has grown in size. The NDVI was higher in 1990 than it was in 2023. Notably, areas such as Keraniganj, Purbachal New Model Town, and Narayanganj, which were once more vegetated, have experienced a marked decline in NDVI in recent years. When compared with LST maps, these low-NDVI zones correspond closely to urban heat hotspots, reinforcing the inverse relationship between vegetation cover and surface temperature. This pattern supports the conclusion that decreasing vegetation is a major driver of UHI intensification in Dhaka.



**Fig. 4** Spatial distribution of urban land surface temperature and the land surface temperature along Profiles 1 and 2 across Dhaka city for 2018, and 2023

## NDVI of Different Years for Greater Dhaka



**Fig. 5** NDVI of Different years showing the high temperature in City centers (a) NDVI of 1990; (b) NDVI of 2000, (c) NDVI of 2010, (d) NDVI of 2017, and (e) NDVI of 2023

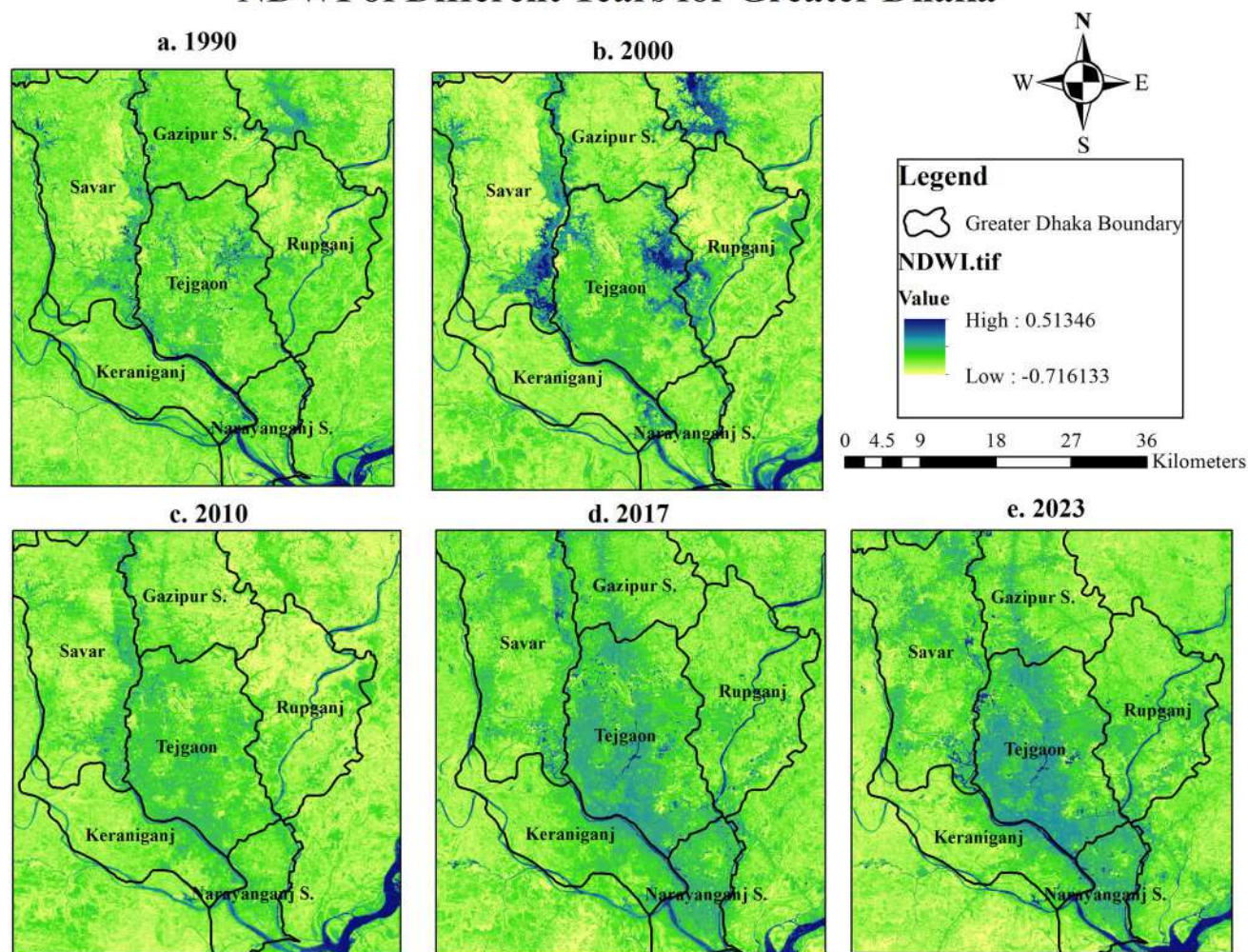
Figure 6 shows visual depictions of the yearly NDWI maps from 1990, 2000, 2010, 2017, and 2023. The blue shade represents high NDWI values. A thorough examination of the photos reveals an apparent reduction in the water level inside the metropolitan region of Dhaka. The NDWI has declined as the metropolitan region has expanded—the maps from 1990 to 2000 show that the NDWI was lower than in 2023. Although some urbanized zones still show NDWI values similar to water bodies, these are often residual or misclassified pixels, not actual water features. The decline in NDWI aligns with areas of rising LST, supporting the conclusion that diminishing water coverage contributes to the UHI effect. Overall, the observed reduction in surface water is a key factor behind the increasing temperature trends in Dhaka's urban landscape.

This image shows the 2008, 2010, 2014, 2017, and 2023 NDBI maps in Fig. 7. The red color indicates a high NDBI. The photos reveal that Dhaka's metropolitan area has grown

significantly, benefiting the NDBI. Maps from 2008 show the NDBI was lower then and higher in 2023. As we observe on the map for 2023, the NDBI has significantly expanded in Dhaka North and South City Corporation, comprising Gazipur, Narayanganj, Keraniganj, Savar, and Rupganj. The progression of NDBI patterns highlights how built-up expansion has replaced vegetation and water bodies, contributing to elevated land surface temperatures. When compared with LST maps (Fig. 3), it is evident that areas with high NDBI also correspond to urban heat hotspots, reinforcing the link between built-up density and the intensification of UHI effects across the Dhaka region.

Annual UI maps for 1990, 2000, 2010, 2017, and 2023 are shown in Fig. 8. Brown indicates high UI values. A close look at the photos shows that Dhaka's Urban Index has risen. The UI grew with urbanization. Over time, cities lose flora and water. The user interface mirrors NDBI. Maps from 2008 show a substantially lower UI than in 2023. The

## NDWI of Different Years for Greater Dhaka



**Fig. 6** NDWI of Different years showing the high temperature in City centers (a) NDWI of 1990; (b) NDWI of 2000, (c) NDWI of 2010, (d) NDWI of 2017, and (e) NDWI of 2023

2023 map shows the UI spreading to Gazipur, Narayanganj, Keraniganj, Savar, and Rupganj in the Dhaka North and South City Corporations. This trend mirrors the growth observed in NDBI maps, further confirming that increased urban density is a major driver of UHI. The expanding UI highlights the spatial footprint of urbanization and underscores its role in altering the microclimate of the greater Dhaka region.

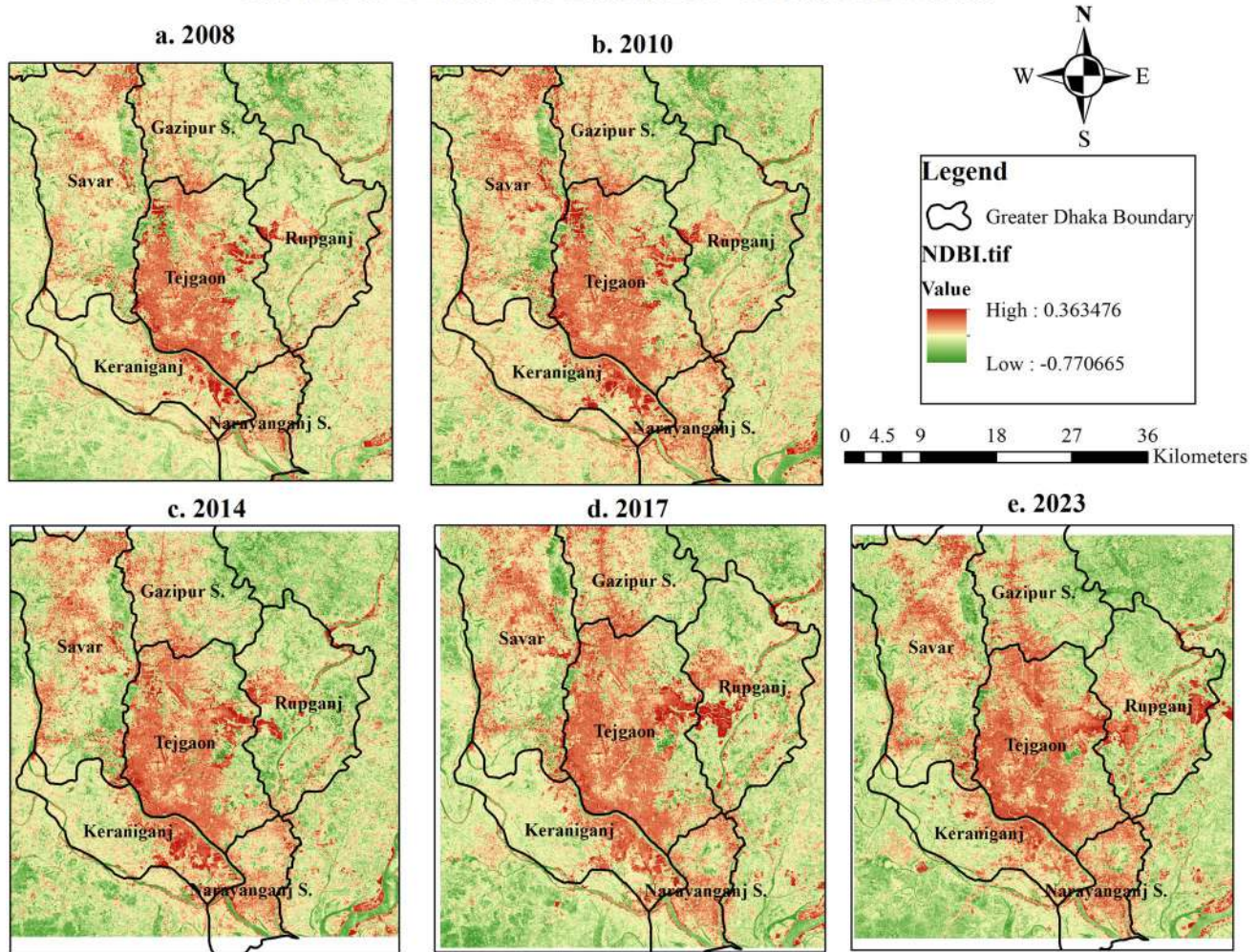
Figure 9 shows the albedo estimations for 2014, 2017, 2020, and 2023, as obtained from Landsat 8. The ratio of reflected to incoming light is known as albedo, and it can be seen that it is increasing with time, with yellow indicating low albedo and blue indicating high albedo levels. In recent years, urbanization has increased the visibility of reflections. As the figure demonstrates, new urban areas have a higher albedo value than older ones. The comparison between 2014 and 2023 highlights a notable rise in overall albedo values, likely resulting from the expansion of

the city and the use of more reflective construction materials. Higher albedo surfaces absorb less heat and contribute to reduced surface temperatures, making them a valuable strategy in mitigating UHI effects. Given the inevitability of continued urban growth, promoting the use of high-albedo materials in urban planning can serve as an effective passive cooling measure.

### 5.4 Hot and Cool Spot Identification

This study identified several areas undergoing significant land use transformation using historical imagery from Google Earth Pro. Figure 10 illustrates the spatial distribution of LST over Dhaka, highlighting significant changes in thermal patterns due to urban expansion. The left panel shows cooler zones in blue, while the right panel displays emerging urban heat hotspots in red. These hotspots

## NDBI of Different Years for Greater Dhaka



**Fig. 7** NDBI of Different years showing the high temperature in City centers. (a) NDBI of 2008; (b) NDBI of 2010, (c) NDBI of 2014, (d) NDBI of 2017, and (e) NDBI of 2023

represent elevated surface temperatures, which have become more pronounced over time due to rapid urban development.

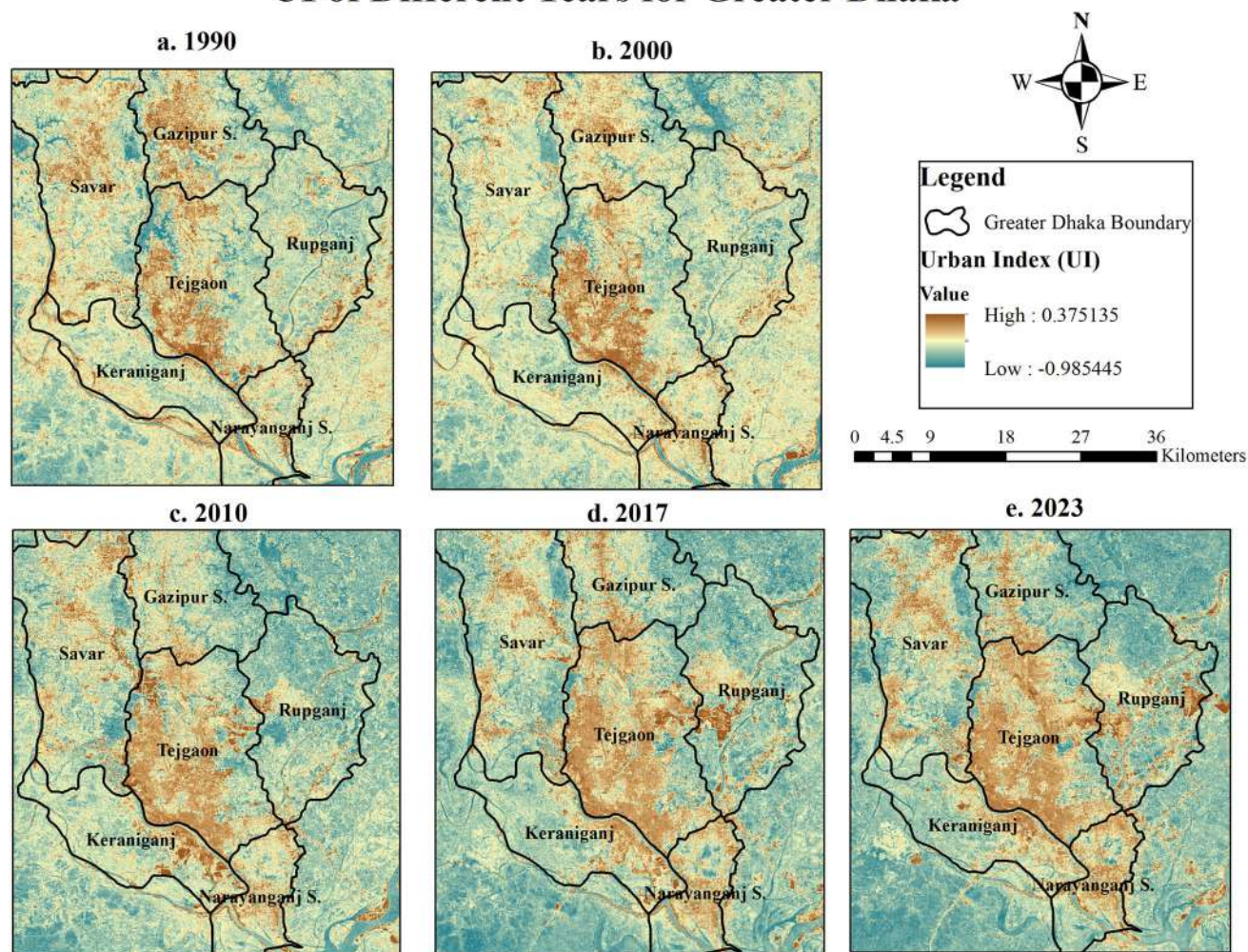
Naturally cooler zones such as Hatirjheel, Chandrima Uddan, Dhanmondi Lake, and other urban green and water-covered areas remain distinctly cooler than their surroundings. These areas act as cool islands amidst the growing UHI effect. The figure reveals that new heat-prone zones have emerged in response to converting open lands, low-lying wetlands, and vegetated areas into built-up infrastructure to accommodate population growth. These transformations have intensified surface temperatures, especially in densely developed neighborhoods.

The circled regions in the right panel of Fig. 10 show key urbanizing hotspots where impervious surfaces have replaced bare lands, forest edges, and waterbodies. Ground observations confirm the strong correlation between land cover change and localized temperature increase. While some buildings show efforts to incorporate rooftop greenery,

these initiatives appear insufficient in mitigating the UHI effect across the city. The results highlight the urgent need for sustainable urban planning that protects natural land covers, promotes green infrastructure, and restores urban ecosystems to counteract rising urban temperatures.

Supplementary Table S-2 lists the hot and cool spots across Greater Dhaka City. Cool spots are primarily associated with natural and vegetated areas, such as Mirpur Lake, the Buriganga River, Osmani Uddan, Chandrima Uddan, Tejgaon Airport, the National Botanical Garden, and Ramna Park. In contrast, older hot spots are concentrated in established residential, commercial, and industrial zones, including Mirpur, Uttara, Tejgaon, Dhanmondi, and Narayanganj Sadar. Notably, new hot spots have emerged in rapidly urbanizing zones such as Kamar Kanda, Jalshiri Abashon, Purbachal, Araihasar Economic Zone, Sonargaon, and Demra, reflecting the impacts of recent economic and residential developments.

## UI of Different Years for Greater Dhaka



**Fig. 8** Urban Index (UI) of Different years showing the high temperature in City centers (a) UI of 1990; (b) UI of 2000, (c) UI of 2010, (d) UI of 2017, and (e) UI of 2023

Supplementary Figure S4 provides field verification at one location. In 2008, the area was relatively undeveloped with open land. In contrast, by 2023, extensive urban construction had transformed it into a newly identified LST hotspot, emphasizing the direct link between land cover change and urban warming.

## 5.5 Model Performance

### 5.5.1 Factors Influencing

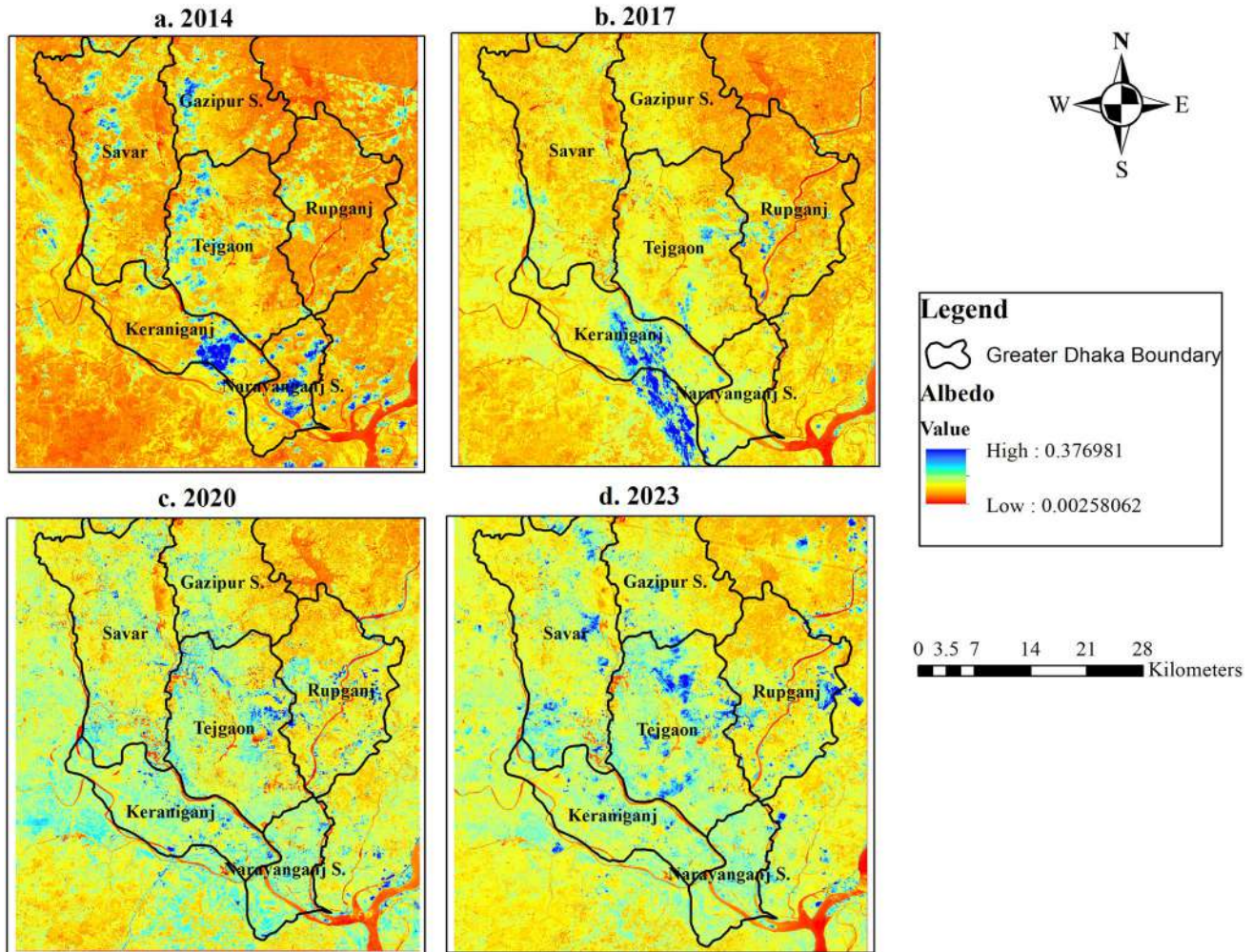
Figure 11 presents a correlation matrix highlighting the relationship between LST and five key indicators: Albedo, NDVI, NDBI, NDWI, and UI. The correlation values range from  $-1$  to  $+1$ , with the color gradient indicating the strength and direction of the relationship—blue for negative correlations and red for positive correlations. The matrix reveals a

largely linear association between LST and the other variables, validating the assumption of direct influence.

Among the indicators, NDVI and Albedo exhibit the strongest negative correlations with LST, at  $-0.4$  and  $-0.3$ , respectively. This aligns with well-established environmental principles: vegetated areas reduce surface temperatures through evapotranspiration, while surfaces with higher reflectivity absorb less solar radiation, resulting in cooler conditions. These findings support the significance of urban greening and reflective materials in reducing UHI intensity.

Conversely, NDBI and UI show strong positive correlations with LST, both at  $0.6$ , indicating that increased built-up intensity and urban development are associated with higher land surface temperatures. This confirms that impervious surfaces and densely built environments significantly contribute to the urban heat island effect. The positive correlation of NDWI ( $0.1$ ) with LST is minimal, suggesting a

## Albedo of Different Years for Greater Dhaka



**Fig. 9** Albedo for Surface Reflectance of Different Years in City Centers (a) Albedo of 2014; (b) Albedo of 2017, (c) Albedo of 2020, and (d) Albedo of 2023

weak or negligible influence of surface water presence on temperature in this context.

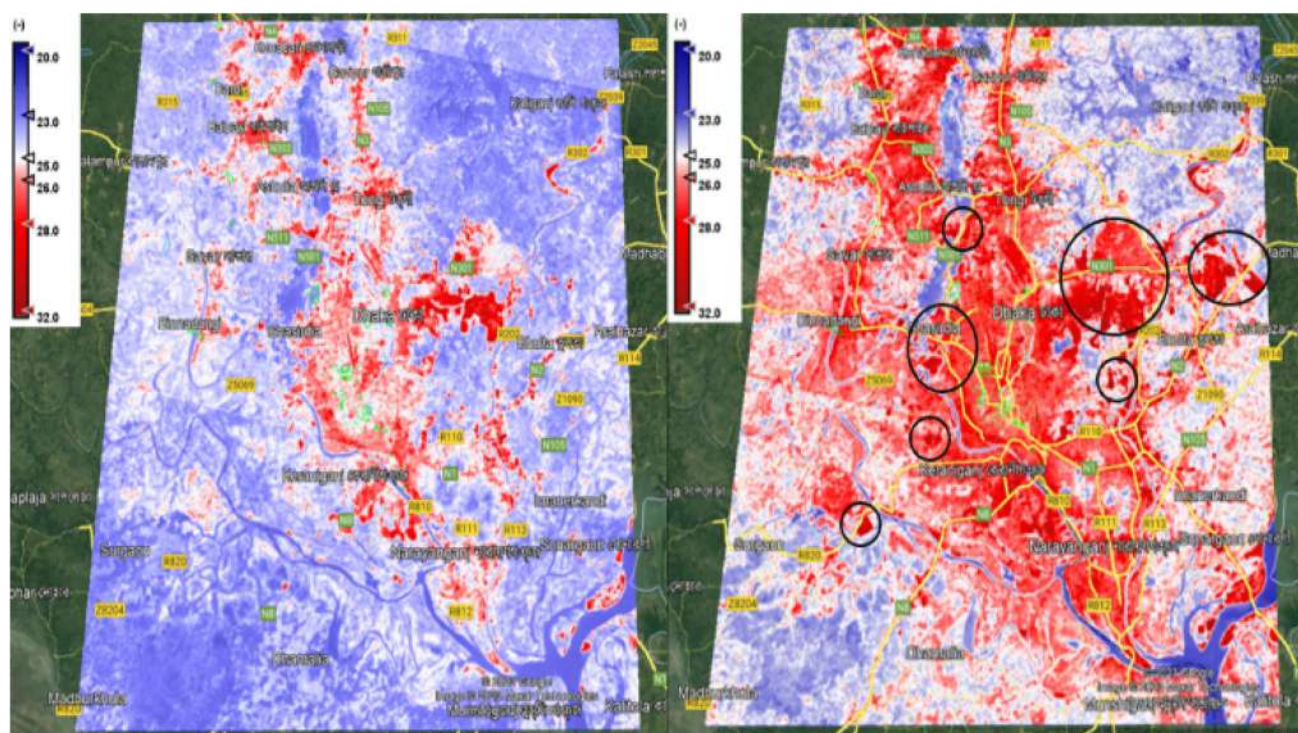
The matrix also shows strong interdependencies among the predictors themselves. For instance, NDVI is negatively correlated with NDBI (−0.5) and UI (−0.5), which is logical, as increased urbanization typically comes at the cost of vegetation loss. Similarly, NDWI and NDBI exhibit a strong negative relationship (−1.0), indicating an inverse spatial relationship between built-up areas and surface moisture. Overall, urbanization indicators (UI and NDBI) are the strongest contributors to increasing LST, while vegetation and surface reflectivity (NDVI and Albedo) are critical for mitigating heat.

These correlations not only validate the model inputs used in this study but also provide further justification for targeting urban greening and reflective surface strategies as part of UHI mitigation. The matrix reinforces that increased

urbanization at the expense of vegetation and water bodies is a key driver of rising land surface temperatures in Dhaka.

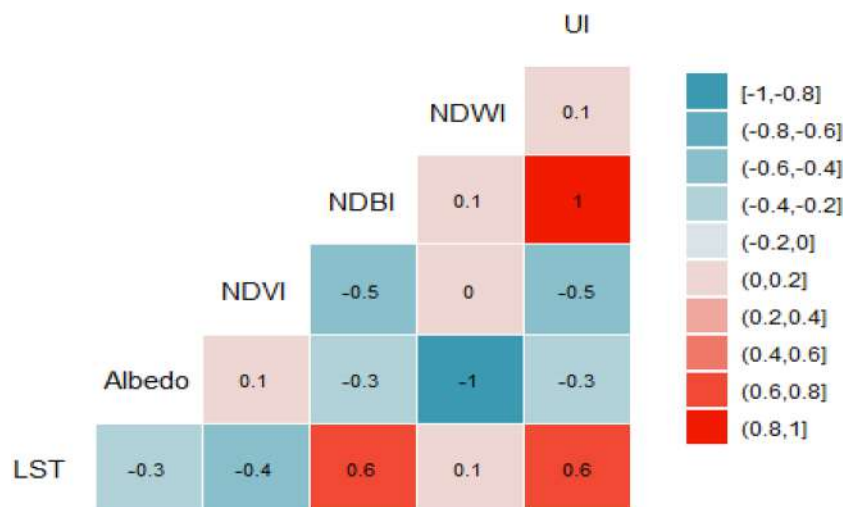
### 5.5.2 Model Performance

The performance of the four predictive models was assessed using RMSE, PBIAS, rSD,  $R^2$  and KGE, as presented in Table 2. Among the models, the SVM showed the best overall performance. It achieved the lowest RMSE value of 0.78, indicating the highest prediction accuracy. In contrast, the linear regression model yielded the highest RMSE (1.00), highlighting its relatively poor predictive capacity. Similarly, in terms of PBIAS, which measures the average tendency of the predicted values to be larger or smaller than their observed counterparts, SVM recorded the lowest bias (1.0%), while linear regression exhibited the highest bias (2.6%).



**Fig. 10** LST of 2016 (Left) and 2023 (Right) in Greater Dhaka City

**Fig. 11** Correlation matrix plot showing the relation of LST with different influencing factors



**Table 2** The performance of machine learning models in predicting LST based on statistical metrics

| Model Evaluation Metrics                      | LinReg | SVM  | RF   | ANN  |
|-----------------------------------------------|--------|------|------|------|
| RMSE (Root Mean Squared Error)                | 1      | 0.78 | 0.85 | 0.87 |
| PBIAS % (Percentage of Bias)                  | 2.6    | 1    | 1.4  | 1.7  |
| rSD (Residual Standard Deviation)             | 1.38   | 1.24 | 1.26 | 1.23 |
| R <sup>2</sup> (coefficient of determination) | 0.69   | 0.7  | 0.68 | 0.69 |
| KGE (Kling-Gupta efficiency)                  | 0.58   | 0.71 | 0.68 | 0.72 |

The rSD further supports the better performance of the SVM and ANN models, which recorded values of 1.24 and 1.23, respectively—lower than linear regression (1.38). These findings indicate that non-linear models are better at capturing the underlying variability in the data.

When considering  $R^2$ , all models demonstrated comparable performance, with values ranging between 0.68 and 0.70. SVM recorded the highest  $R^2$  value (0.70), slightly outperforming the other models. The KGE, a more integrative performance metric that incorporates correlation, bias, and variability, was also highest for the ANN (0.72), followed closely by SVM (0.71). This further confirms the

effectiveness of non-linear machine learning models in replicating the observed patterns.

Results indicate that while all models showed acceptable performance, the SVM model consistently outperformed the others across most evaluation metrics, indicating it is the most reliable method for the current application. Linear regression, on the other hand, demonstrated the weakest performance, particularly in terms of error magnitude and bias.

The results reveal that while all models provided reasonable predictive performance, SVM consistently outperformed the others across most evaluation criteria, making it the most robust and reliable model for UHI prediction in this study. The relatively poor performance of the linear regression model highlights its limitations in handling the non-linear and multifactorial nature of urban climate dynamics.

### 5.5.3 Model Predicted LST

Figure 12 compares model-predicted LSTs with the observed Landsat 8 LST for 2023. The Landsat 8 image is the reference dataset, showing a detailed spatial distribution of LST values across the Dhaka Metropolitan Area, with temperatures ranging from approximately 19 °C to 35 °C. The red zones indicate urban hotspots with higher LSTs, while green and yellow areas represent cooler vegetated or peri-urban zones.

The LM significantly underperforms compared to the reference. It shows a smoother and overly generalized temperature pattern with diminished spatial contrast. The urban heat signatures captured in the Landsat 8 image are less pronounced, and many high-temperature zones are underestimated. This indicates the linear model's limited capability to capture the complex, nonlinear interactions influencing urban LST variability.

In contrast, the SVM, RF, and ANN models more accurately replicate the spatial variability of LSTs observed in the Landsat 8 image. These machine learning models capture the hot zones in central Dhaka, especially around high-density built-up areas, with greater fidelity. In particular, the SVM and ANN models show higher visual alignment in red-hot areas, suggesting their superior ability to detect and map thermal patterns within heterogeneous urban environments. The RF also performs relatively well, capturing many core heat zones, but slightly less intensely than ANN and SVM. The spatial distribution of LSTs predicted by RF still maintains a good correlation with the reference, indicating robustness in handling spatial heterogeneity.

In summary, the visual comparison clearly demonstrates that SVM and ANN outperform LM and even RF in reproducing the complex urban heat landscape seen in the Landsat 8 image. Their performance underscores the effectiveness

of nonlinear machine learning models in urban climatology, particularly for mapping UHI patterns. These results support the recommendation to adopt SVM and ANN for more accurate and reliable LST predictions in future urban climate studies.

### 5.5.4 Model Prediction Bias

Figure 13 presents the spatial distribution of bias in temperature predictions across four modeling approaches: LM, SVM, RF, and ANN. The bias maps depict underestimation and overestimation patterns, where green shades indicate negative bias and brown shades reflect positive bias. Ideally, model accuracy improves as the bias approaches zero, signifying minimal deviation from actual observations. The LM model displays the most widespread and pronounced bias, with strong positive deviations particularly evident in urban cores. In contrast, the SVM and ANN models show considerably lower bias across the study area, with more uniform distribution and minimal extremes, suggesting better model generalization. The RF model performs moderately well but exhibits slightly more spatial variability in bias than SVM and ANN.

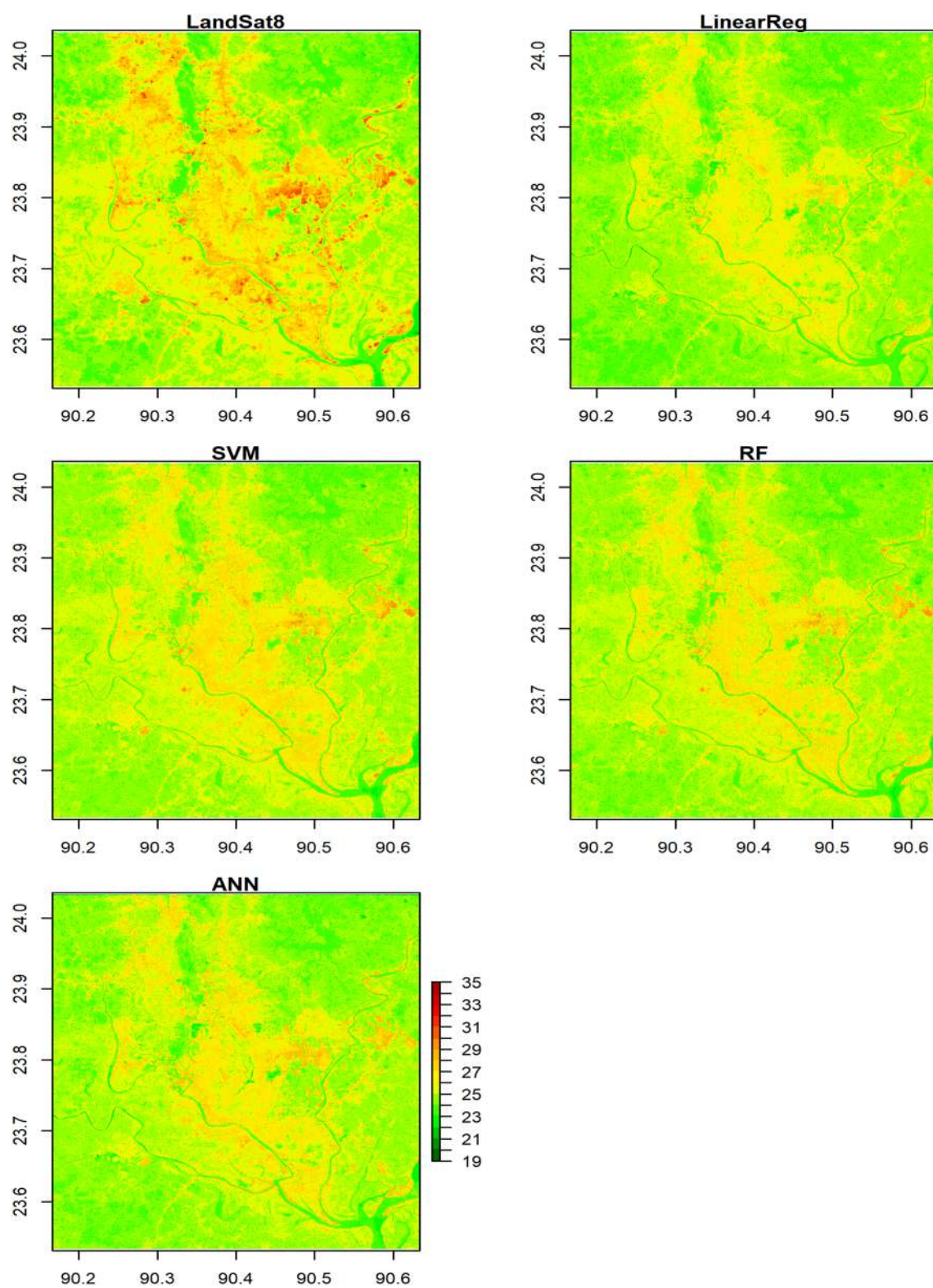
Among all models, SVM and ANN demonstrate the most accurate and balanced performance, with minimal spatial bias and stronger predictive stability. These findings support the recommendation that machine learning techniques, particularly SVM and ANN, are more suitable for mapping and predicting land surface temperature due to their robustness in handling nonlinear relationships and complex urban patterns.

### 5.5.5 Sensitivity Analysis

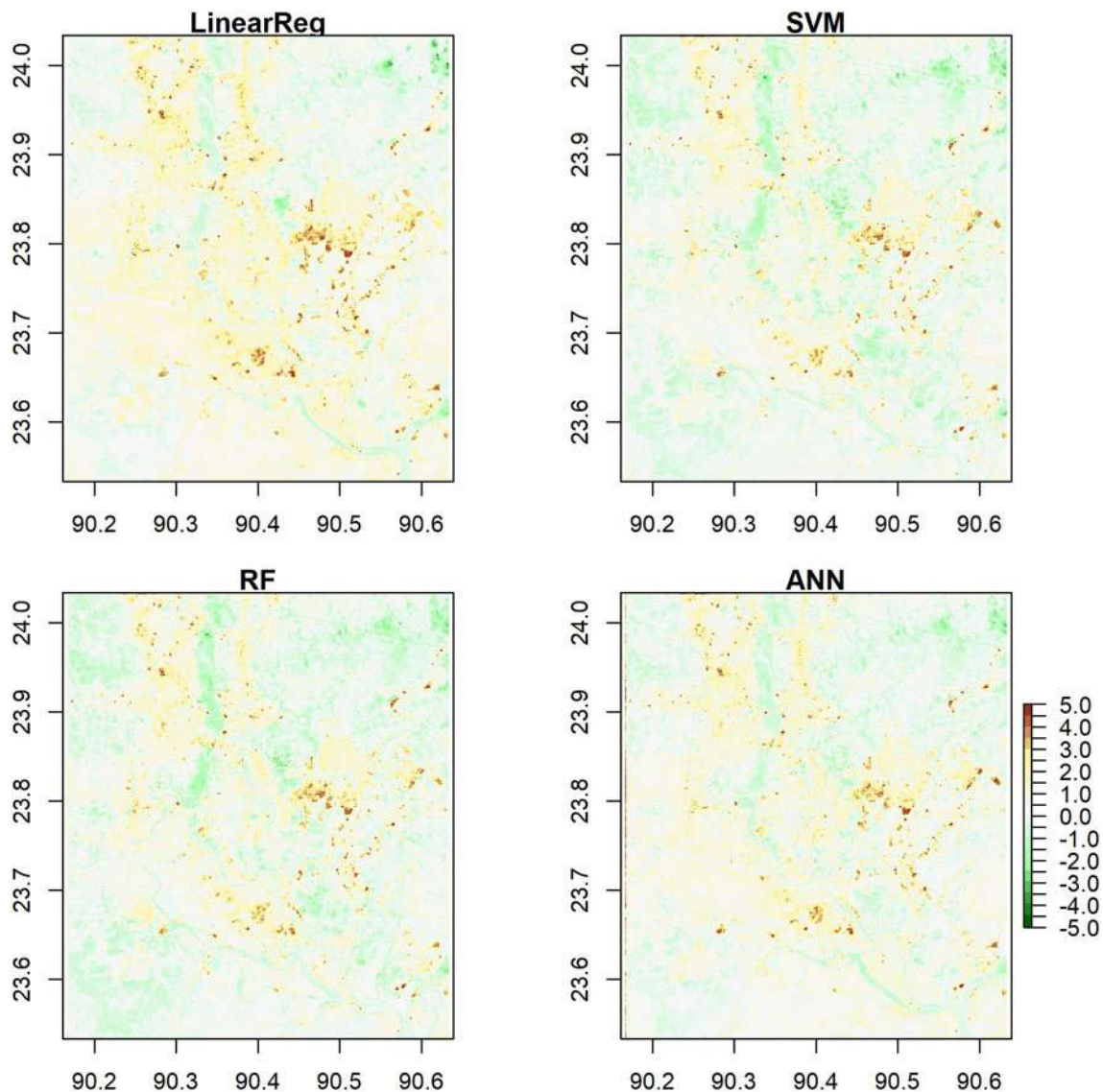
As illustrated in Fig. 14, a  $\pm 25\%$  variation in LST was considered to capture the model's sensitivity to changes in key urban and environmental indices. This sensitivity analysis provides valuable insights into how different predictors influence urban thermal dynamics proportionally.

The graph's red and purple buffered lines represent the Urban Index (UI) and the Normalized Difference Built-up Index (NDBI), respectively, showing a strong positive correlation with LST. An increase in either index leads to a corresponding rise in urban temperatures. Notably, a 10% reduction in UI could reduce LST by approximately 0.5 °C. Although large-scale de-urbanization is unlikely, the model suggests a 10% decline in NDBI could reduce LST by up to 2 °C. These findings emphasize the thermal burden of expanding impervious surfaces and dense urban infrastructure.

Conversely, predictors such as NDVI, NDWI, and Albedo are inversely related to LST. The khaki-colored line



**Fig. 12** Spatial distribution of predicted LST using different machine learning models



**Fig. 13** Bias in LST prediction using different machine learning models

representing Albedo demonstrates its strong cooling influence—higher surface reflectivity significantly reduces thermal absorption, lowering surface temperatures. However, implementing high-albedo materials in large-scale urban settings may pose financial and logistical challenges. Still, the design of taller buildings with reflective surfaces can contribute to passive cooling by minimizing solar energy absorption.

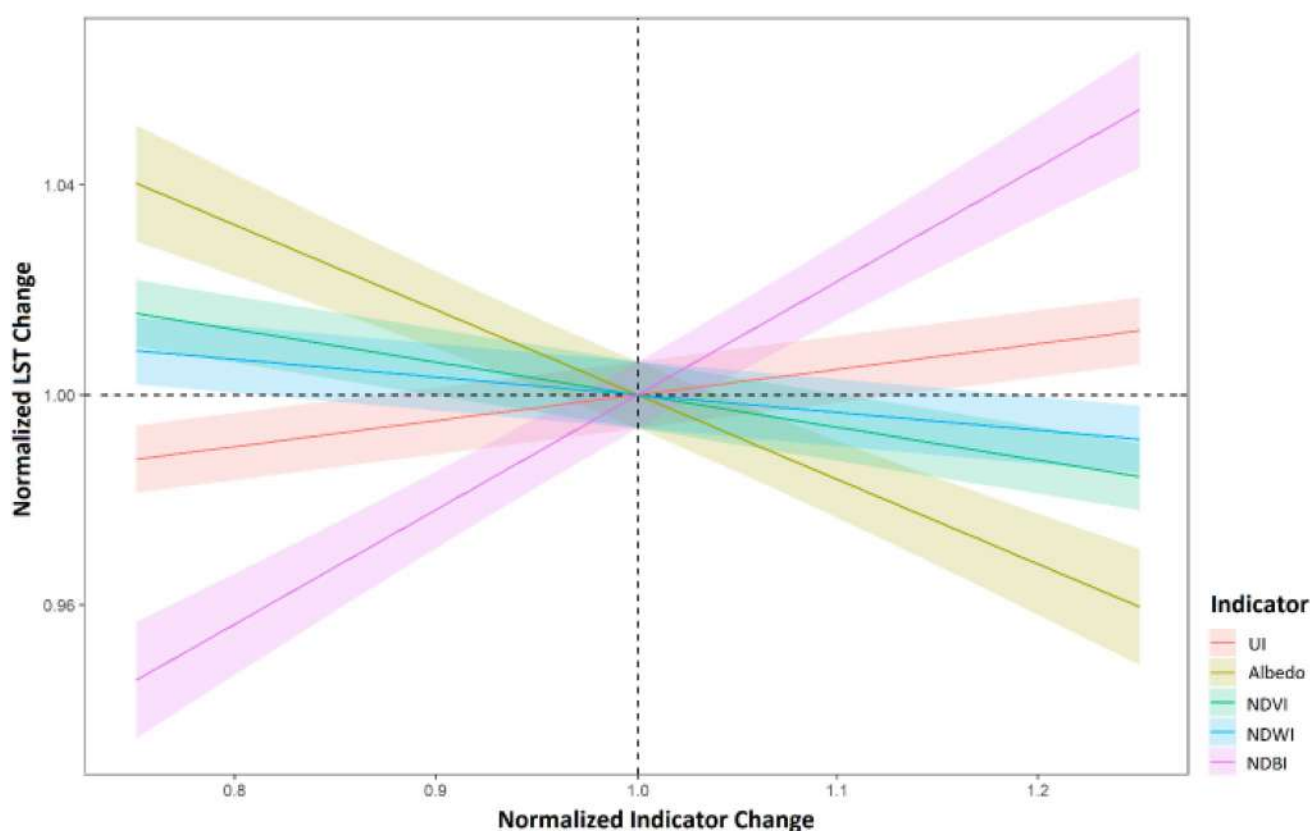
The green and blue buffered lines represent NDVI and NDWI, both of which display a substantial cooling effect. According to the model, a 20% increase in NDVI and NDWI can reduce LST by approximately 12% and 7%, respectively. More impressively, a 25% increase in NDVI alone may lead to a 15% reduction in LST, equating to a 2–3 °C drop in ambient temperature across the Dhaka Metropolitan Area. These results highlight the critical importance of

urban greening and water-sensitive urban design in mitigating UHI effects.

## 5.6 Suitable Mitigation Measures for DMA

The recommended mitigation measures for Greater Dhaka have been carefully selected based on their relevance, comprehensibility, controllability, and sensitivity to the local context. Dhaka, characterized by rapid unplanned urban growth, high population density, and limited green infrastructure, is highly vulnerable to UHI effects (Kafy 2021; Rahman et al. 2022a, b).

Cool pavements and roofs, such as reflective and green surfaces, are particularly suitable as they can be implemented on existing infrastructure and significantly reduce surface temperatures (Santamouris 2013; Mills et al. 2021).



**Fig. 14** Sensitivity of the model showing normalized LST change based on the predictor change

Given Dhaka's dense built environment and high solar exposure, these materials offer a cost-effective and scalable solution for immediate temperature reduction. Increasing urban vegetation through street trees, vertical greenery, and green walls is crucial in a city where natural landscapes are being rapidly lost to development (Bowler et al. 2010; Mircholi et al. 2020). These nature-based solutions cool the ambient air and enhance urban biodiversity and air quality (Gunawardena et al. 2017). Urban forestry and green spaces are critical thermal buffers in densely populated neighborhoods (Pervin et al. 2022).

Urban planning interventions, including land use management, open space preservation, and climate-responsive design, are vital for steering Dhaka toward sustainable urban growth (Rahman et al. 2022a, b; Seto et al. 2021). Integrating green corridors and open spaces into future urban development can help balance density with environmental health (Kabisch et al. 2016). Water body management, such as protecting existing water bodies and creating new ones, helps restore aquatic ecosystems' natural cooling function (Li et al. 2023). This is especially relevant for Dhaka, where rivers and wetlands are rapidly shrinking (Nuruzzaman 2015). Building material modifications, including high-albedo materials and appropriate building geometry, can reduce heat absorption and improve thermal comfort (Santamouris

2013; Pour et al. 2022). These strategies are practical for new construction and retrofitting existing buildings in Dhaka's mixed urban fabric.

Lastly, community engagement, public awareness, and reduced motor vehicle use are essential behavioral strategies. In a city with intense congestion and limited enforcement capacity, empowering communities to participate in green initiatives and promoting non-motorized transport can support broader UHI mitigation goals (Maheng et al. 2019; Rahman et al. 2022a, b).

Collectively, these measures are technically viable for Dhaka and adaptable to its socio-economic and climatic realities. Their implementation can significantly enhance urban livability and climate resilience in the face of rising temperatures Table 3.

## 6 Discussion

This study utilized Landsat-derived LST data and machine learning models to map and quantify the spatiotemporal dynamics of the UHI effect in Dhaka. A multi-year analysis (1990–2023) revealed the emergence and intensification of UHI hotspots, reflecting the influence of rapid urban expansion, reduced vegetation cover, and changing land use.

**Table 3** Possible mitigation measures based on the four criteria

| Mitigation Measures                                                                         |                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cool Pavement<br>(Santamouris 2013; Mills et al. 2021; Pour et al. 2022)                    | Reflective pavement<br>Green pavements<br>Pervious pavements                                                                                                                    |
| Cool Roof                                                                                   | Reflective roofs<br>Green roofs                                                                                                                                                 |
| Increase Urban Vegetation (Bowler et al. 2010; Gunawardena et al. 2017; Pervin et al. 2022) | Planting street trees<br>Preserving and increasing green space<br>Green wall<br>Urban forestry<br>Vertical greeneries                                                           |
| Urban Planning (Rahman et al. 2022a, b; Seto et al. 2021; Kabisch et al. 2016)              | Change of land cover<br>Increasing open spaces and Greenery<br>Urban planning and land use management<br>Planned urban development<br>Climate-Responsive Design                 |
| Water Body (Nuruz-zaman 2015; Li et al. 2023)                                               | Groundwater table management<br>Existence of water bodies<br>New waterbody generation                                                                                           |
| Building Materials (Santamouris 2013; Pour et al. 2022)                                     | Modifying building geometry and materials<br>Building mass, construction materials, and surface characteristics:<br>High albedo materials and pavements<br>Green Infrastructure |
| Public Awareness (Maheng et al. 2019; Rahman et al. 2022a, b)                               | Public and community Engagement<br>Population management<br>Public awareness<br>Reducing motor vehicle use                                                                      |

These findings align with recent research, such as Trotter et al. (2022) and Kafy et al. (2021), who observed that Dhaka's peri-urban areas are becoming increasingly vulnerable due to unplanned development. The accelerating pace of urbanization, coupled with inadequate planning and enforcement, has dramatically increased the UHI effect in these areas, creating a feedback loop of heat retention and infrastructure stress (Khatun et al. 2023). The results obtained in the study are discussed in the following subsections.

## 6.1 Urbanization and LST Patterns

The annual and seasonal LST analyses clearly demonstrate the increasing trend of surface temperatures across the Dhaka Metropolitan Area, with multiple new UHI hotspots forming over time. Areas such as Tongi, Gazipur, Savar, and Narayanganj experienced notable warming between 1990 and 2023, particularly around the Buriganga River, where urban development has significantly intensified. This uncontrolled urban growth, particularly along the Buriganga and other riverbanks, has made the region more susceptible to extreme heat, as these areas were once cooling zones (Rahman et al. 2021). Seasonal LST assessments showed that wintertime temperatures also rose markedly between 2013 and 2020, with the 2020 record high likely influenced by

climatic extremes and anthropogenic heat emissions. Pre-monsoon and post-monsoon periods reflected similar warming trends, with 2023 standing out for experiencing multiple heatwaves and widespread thermal discomfort. The more frequent and intense heatwaves are likely the result of both climate change and the exacerbation of urban sprawl, which traps heat and prevents its natural dissipation (Singh et al. 2022). The observed increasing trend in annual and seasonal LST mirrors findings from other major South Asian cities. Adnan et al. (2023) reported that Karachi, Pakistan, has witnessed similar seasonal LST increases tied to peri-urban sprawl. During the pre-monsoon, high humidity traps heat, making it hard for the city to cool at night. Combined with air pollution, this creates a "heat trap," raising temperatures and worsening air quality (Khan et al. 2020a; Rahman and Hossain 2021). In Dhaka, new UHI hotspots emerged in areas like Tongi, Gazipur, Savar, and Narayanganj, consistent with Mahmood et al. (2022), who emphasized peri-urban transformation as a primary UHI driver. Monsoon UHI is not calculated due to the continuous data availability of Landsat images.

The UHI profile analysis further confirmed that built-up urban regions consistently exhibited the highest temperatures, while vegetated, peri-urban, and water-rich zones (e.g., Dhanmondi Lake, Hatirjheel) remained comparatively cooler. Interestingly, Dhaka's urban heat islands are more pronounced near areas lacking green cover and water bodies, which traditionally help in mitigating urban heat. The absence of these natural cooling agents further exacerbates the situation (Akbari et al. 2009). The impact of the COVID-19 lockdown in 2020 was evident in a temporary dip in LST, likely due to reduced industrial and transportation activities. However, this cooling effect dissipated by 2023, as newly developed peri-urban zones also exhibited higher temperatures. Notably, the COVID-19 lockdown-linked cooling observed temporarily in 2020 aligns with global observations, such as in Wang et al. (2022) for cities in China. Although the short-term reductions in emissions during the lockdown led to lower temperatures, the long-term trends reveal the unsustainable nature of such measures without systemic urban changes.

This study addressed intra-annual variability by conducting a seasonal analysis of LST across pre-monsoon, post-monsoon, and winter periods. However, diurnal variability was not analyzed due to the temporal limitations of the Landsat satellite, which offers imagery at a 16-day interval and lacks sub-daily resolution. Despite this limitation, it is well-recognized that LST in Dhaka can vary substantially within a single day. For instance, during the pre-monsoon season, high humidity levels hinder nocturnal cooling, intensifying nighttime heat stress. This phenomenon, compounded by elevated levels of air pollution, creates

a persistent “heat trap” effect that exacerbates both thermal discomfort and air quality issues (Khan et al. 2020a; Rahman and Hossain 2021).

## 6.2 Biophysical Indicators and Urban Expansion

Vegetation indices, particularly NDVI, consistently declined over time, especially in areas like Keraniganj and Purbachal New Model Town, indicating vegetation loss due to urban encroachment. Conversely, NDBI and UI values increased, signifying expansion of impervious surfaces and built-up areas. This shift in land use contributes to the amplification of the UHI effect. Albedo analysis showed an increase over the study period, suggesting that newer urban materials may reflect more solar radiation; however, this may be offset by surface roughness and lack of vegetation, which compromise the cooling potential. While albedo has improved with the introduction of reflective surfaces, these improvements have been minimal due to the continued loss of green spaces, which are essential for cooling urban environments naturally (Mircholi et al. 2020).

Declines in NDVI and NDWI, alongside increases in NDBI and UI, confirm that Dhaka’s vegetation and water bodies are being rapidly lost to urban development. Similar patterns were documented by Pervin et al. (2022) in Chattogram and by Mircholi et al. (2020) in Tehran, where impervious surface growth significantly raised urban temperatures. The reduction in water bodies in the city has compounded the problem, as these bodies naturally help regulate temperatures, especially during the dry season (Rahman et al. 2021). The observed partial increase in albedo likely reflects the mixed urbanization trend: while some modern building materials have higher reflectivity, widespread vegetation loss counterbalances these benefits. The present study’s emphasis on spatial mapping of biophysical indicators over multiple decades strengthens the regional understanding of UHI drivers compared to earlier studies that mainly focused on short periods (e.g., Uddin et al. 2022).

## 6.3 Predictive Modeling and Evaluation

To mitigate UHI effects in Dhaka, strategies from global best practices can be applied. Increasing green spaces, cool roofing, and water management help reduce temperatures (Akbari et al. 2001). Urban planning should focus on mixed-use development and tree planting to provide shade. Public education and energy efficiency improvements can also reduce UHI (Gómez-Baggethun and Barton 2013). Implementing heat action plans, as in Ahmedabad, can help manage heat risks (Mishra et al. 2012). These globally proven strategies offer a foundation for Dhaka’s unique challenges.

Machine learning models, including SVM, RF, and ANN, were applied to enhance UHI prediction accuracy by utilizing multiple urban and environmental indicators. Among the models, ANN and SVM demonstrated superior performance across most evaluation metrics, including RMSE, PBIAS, and KGE. These models outperformed linear regression, which showed the highest error and bias. The KGE values, ranging around 0.7 for ANN and SVM, confirm their suitability for modeling complex non-linear interactions in urban climate systems. Pour et al. (2022) demonstrated that SVM and ANN outperform traditional regression in capturing nonlinear LST patterns in Kuala Lumpur and Jakarta. Your use of multiple metrics (RMSE, KGE, rSD) provides a comprehensive validation, unlike some older studies that relied solely on  $R^2$  values.

Model sensitivity analysis further illustrated the relative influence of various predictors on LST. NDVI and NDWI were found to be the most effective in reducing surface temperature, with a 25% increase in NDVI capable of reducing LST by up to 2–3 °C. Conversely, increases in UI and NDBI were associated with significant rises in LST, highlighting the thermal burden of urban expansion. This model sensitivity analysis is crucial for urban planners, as it provides actionable insights into which urban features most significantly influence UHI, enabling targeted interventions (Zhao et al. 2023). Albedo had a notable cooling effect, but implementing high-albedo materials in urban construction may face economic and infrastructural barriers. The strong sensitivity of NDVI and NDWI in your models also echoes findings by Li et al. (2023), who showed that a 20% increase in vegetation could lower urban LST by up to 3 °C in Delhi and Mumbai. The documented role of Albedo improvement, while secondary to vegetation increase, is consistent with Rahman et al. (2022a, b), who found cool roof initiatives modestly lower LST in Dhaka’s core zones.

## 6.4 Implications for Urban Planning

These findings emphasize the urgent need for sustainable urban planning practices in Dhaka. Strategic interventions, such as enhancing green infrastructure through vertical greening, rooftop gardens, and urban forestry, can help mitigate UHI impacts. Water-sensitive urban design and the use of reflective construction materials also offer potential benefits. However, achieving large-scale UHI mitigation requires collaboration between government agencies, private developers, and local communities to overcome land-use barriers and achieve urban resilience (Mircholi et al. 2020). Dhaka’s high population density and rapid land-use transformation limit the scalability of specific solutions, such as large-scale greening. Urban greening, reflective surfaces, and water-sensitive planning have also been

emphasized in Seto et al. (2021) for Southeast Asian megacities. However, this study's detailed quantification (e.g., a 25% NDVI increase potentially reducing LST by 2–3 °C) provides practical targets for planners, which are often lacking in prior UHI mitigation studies in Bangladesh.

In addition to physical and environmental interventions, implementing UHI mitigation strategies in Dhaka must contend with significant socio-political barriers. Unclear land tenure in many informal settlements poses a major obstacle to investing in long-term resilience measures such as green infrastructure and community-managed open spaces (Ali et al. 2018). Moreover, limited and inconsistent funding—both domestically and from international sources—restricts the scalability of solutions like cool roofing, green corridors, or urban forestry (Ahmed & Hasan 2012; Ahmed et al. 2013). Trade-offs also exist: while increasing urban greenery is essential for lowering surface temperatures, it requires substantial water input—a resource already under pressure in Dhaka's water-stressed context (Khan et al. 2020b). Balancing water-sensitive urban design with equitable distribution of green spaces thus becomes a critical planning challenge (Chakraborty et al. 2021). Overcoming these barriers requires integrated governance approaches, equitable land policies, and financial mechanisms such as public-private partnerships and climate resilience funds.

Continued urban monitoring, utilizing machine learning models and remote sensing, can support data-driven decision-making. Moreover, integrated with municipal planning, public awareness campaigns can facilitate climate-resilient urban development in Bangladesh's rapidly urbanizing regions. These mitigation proposals align well with recent global best practices. Moreover, the emphasis on machine learning-supported monitoring for future urban management makes your study forward-looking. Zhao et al. (2023) advocate for ML-driven predictive mapping to design climate-resilient cities under SSP-based climate scenarios, a direction your study already initiates. The integration of dynamic urban planning frameworks, supported by continuous remote sensing and data analytics, is pivotal in ensuring long-term UHI mitigation. Ultimately, urban policies must be comprehensive, addressing not only the physical development of the city but also the social and economic factors that influence the effectiveness of climate resilience strategies.

## 7 Conclusions

This study aimed to evaluate the spatiotemporal dynamics of UHI in Dhaka using multi-temporal Landsat satellite data (1990–2023) and machine learning models to quantify the relationship between LST and biophysical indices

such as NDVI, NDBI, NDWI, UI, and Albedo. The analysis revealed that the increase in impervious surfaces and decrease in vegetation significantly intensified UHI effects in Dhaka over time. Machine learning models, particularly SVM and ANN, demonstrated superior performance over traditional linear regression, offering improved LST predictions. Sensitivity analysis showed that NDVI and NDWI have the most substantial cooling effects on LST, while NDBI and UI exacerbate thermal stress. A 25% increase in NDVI alone could reduce urban LST by up to 15%, highlighting the critical role of green infrastructure in urban climate regulation.

The study's key lessons reinforce that natural features, such as rivers, parks, and vegetation corridors, function as urban cool zones. At the same time, densely built-up and poorly ventilated areas act as thermal hotspots. These findings underscore the importance of integrating ecological considerations into city planning to mitigate UHI effects.

Despite its valuable contributions, the study faces several limitations. This study lacks ground-based real-time LST monitoring in Bangladesh, which limits the validation of satellite-derived data with in-situ measurements. Field-based thermal observations were also not conducted due to a mismatch between satellite data timelines and real-time conditions. Consequently, the analysis relies solely on satellite imagery, with field visits used only to verify land cover and the spatial accuracy of hot and cool spot identification. Future research should incorporate additional factors such as humidity, air pollution, and socio-economic dynamics, which significantly influence UHI intensity and distribution. Integrating ground-based temperature data, high-resolution urban morphology, and urban-scale modeling frameworks can enhance the accuracy of mitigation scenario simulations, such as those involving green roofs, reflective materials, and water-sensitive design. Future research should also explore further methods to enhance the accuracy and robustness of UHI prediction using advanced ensemble models, such as XGBoost and deep learning approaches.

Targeted strategies should prioritize vulnerable areas, including green roofs in dense zones such as Motijheel and Gulshan, formalized land tenure in informal settlements like Mirpur and Keraniganj, and community gardens in Old Dhaka, to enhance access to green spaces. Public-private partnerships can help finance resilience projects, particularly in flood-prone areas, while community engagement and awareness campaigns can enhance long-term environmental resilience.

**Supplementary Information** The online version contains supplementary material available at <https://doi.org/10.1007/s41748-025-00810-z>.

**Acknowledgements** The authors appreciate the USGS (United States

Geological Survey), a scientific agency under the US Department of the Interior, for making Landsat data available through online portals. The Ministry of National Science and Technology (NST) of Bangladesh funded the study.

**Author Contributions** All of the authors participated in formulating and creating the research. A.S.M. Maksud Kamal developed the concept; Anika Tabassum collected data; Shamsuddin Shahid and Anika Tabassum conducted the data processing and modeling; Anika Tabassum prepared an initial draft of the paper; and A.S.M. Maksud Kamal, Shamsuddin Shahid, and Md Jakir Hossain revised the article multiple times to produce the final version.

**Data Availability** The research used publicly accessible data. The respective author may also share these resources upon request.

**Code Availability** The programs and codes used for data processing might be shared with the respective author upon request.

## Declarations

**Conflict of interest** The authors assert that they have no conflicting interests.

**Ethics Approval** Morality and honesty are upheld throughout this course. Authorization is taken before snapping photos. The concept and draft texts are human-written, and to improve the sentence structure and grammatical errors, AI tools are used.

**Consent To Participate** Consent from participants is obtained before field observations.

**Publication Consent** All of the authors agreed to publish the research.

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